

The Latent Bridge: A Continuous Channel Between Slow Reasoning and Fast Reactive Models

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Abstract

Real-time interactive AI systems couple deliberative reasoning to reactive action selection through a text prompt (Gemini Live, OpenAI Realtime). We test an alternative: a learned continuous-valued *latent bridge* that projects slow-model residuals into the fast model’s input embedding space, LLaVA-style. We couple a frozen 9B reactive multimodal model to a frozen 8B reasoning model at matched scale on 7 Atari games and a driving domain (MetaDrive). Under a fixed *greedy* action decoder—our original convention—the latent bridge beats the text bridge by 26–82% on 4 of 7 games. We show this advantage is *decoder-specific*: under any fixed sampling temperature it disappears, because the latent’s thin per-emission signal (3–8% variance) improves the deterministic argmax but is swamped once the decoder samples. The honest comparison therefore treats the action decoder as a deployment hyperparameter and evaluates each channel at *its own best decoder*, selected on held-out seeds (leave-one-seed-out) so the estimate is unbiased. Under this best-achievable protocol the latent is **never significantly worse than the text bridge on any of the 7 games and is significantly better on 2** (MsPacman +57%, RoadRunner +28%); the other 5 are statistical ties. The optimal decoder is itself *channel-dependent* (the latent prefers greedy or high temperature, the text bridge $\tau \approx 0.5$), which is exactly why comparing them at a single fixed decoder is unfair to both. The latent is thus a safe-or-better drop-in for the text channel—non-inferior everywhere, materially better where slow reasoning helps most—rather than a uniform winner. *Combining* both channels in one forward pass does *not* help: it never beats the better single channel and significantly *interferes* on 3 of 7 games (catastrophically on RoadRunner, −96%), so the design rule is to couple via exactly *one* channel. The decoder-robust, task-level finding is a behavioral predictor: pooling the 7 Atari games with MetaDrive, latent and text benefits over fast-only track at Pearson $r = 0.92$ —the bridge helps if and only if slow reasoning beats reaction ($T > F$). MetaDrive is a controlled negative where it does not, with a bridge-replacement control confirming the latent is inert there. We also find no bandwidth ceiling (a 30B slow model does not widen the gap; more text does not close it). We release F/T/L replay recordings for all 8 experiments and reproducible pipelines.

Code, data, demos: <https://github.com/bojieli/latent-bridge-games>

1 Introduction

A reasoning model deliberates; a reactive model reacts. Their tick rates differ by an order of magnitude (seconds vs. tens of milliseconds), and the production answer for coupling them is to have the slow model emit text that the fast model reads. We test the alternative: a learned continuous-valued channel between two frozen models.

Setting. Both endpoints are general-purpose multimodal LLMs at matched scale (9B fast, 8B slow), deliberately chosen so the channel (not a capability gap) is the load-bearing variable. The fast model picks actions at ~ 15 Hz; the slow emits at ~ 1 Hz; the bridge is the only learned component. Our domain is Atari [4], where slow/fast coupling is visible to the eye—100 ms ghost-dodging meets 10 s route-planning.

What this paper claims, sharpened.

- **Architecture matters.** A first attempt (v1) using 256-d cross-attention at 2 of 36 LLM layers converged to $KL=0.004$ on offline data but *failed* at deployment. The fix (v2) is the LLaVA pattern: project slow residuals into the fast LLM’s 4096-d input embedding space and prepend, letting all layers attend via standard causal attention.
- **Tuned per channel, the latent is never significantly worse than text and is significantly better on 2 of 7 games** (§4.4, Table 2). Under a fixed greedy decoder the latent beats text by 26–82% on 4 of 7 games (Figure 1)—but *this is greedy-specific*: a full decoder sweep (greedy, $\tau \in \{0.3, 0.5, 0.7, 1.0, 1.5\}$) shows the advantage vanishes at every fixed sampling temperature, because the latent’s thin per-emission signal only sharpens the deterministic argmax (§4.3). Since the action decoder is a deployment hyperparameter one tunes, the fair comparison is each channel at its own best decoder, selected on held-out seeds: there the latent significantly wins MsPacman (+57%) and RoadRunner (+28%), is a statistical tie on the other 5 games, and never significantly loses—and the two channels prefer *different* decoders (latent greedy/high- τ , text $\tau \approx 0.5$), which is why a single fixed decoder is unfair to both.
- **Using both channels at once *hurts*—couple via exactly one** (§4.5, Table 3). Feeding the text suffix and latent tokens to the frozen head in one forward pass never beats the better single channel (0 of 7) and significantly *interferes* on 3 of 7 (MsPacman −49%, River Raid −29%, RoadRunner −96%). The naive “more context is better” intuition fails: the head, trained on one conditioning signal at a time, gets a worse policy from two. The recommendation is to couple via a single channel (latent as the safe default), not both.
- **When it helps is a property of the task, not the channel.** Across 7 Atari games and a non-Atari driving domain (MetaDrive), the latent’s benefit over fast-only ($L-F$) tracks the text bridge’s benefit ($T-F$) at Pearson $r = 0.92$ (and $r = 0.94$ over all 16 evaluated cells, so it does not depend on variant selection): the bridge pays off exactly when slow reasoning beats fast reaction on the task ($T > F$). MetaDrive is a controlled negative—even with a planning-heavy route, $T \leq F$, and a bridge-replacement control confirms the latent is inert. We frame the contribution this way rather than as a “text is bandwidth-limited” claim: our own ablations (a 30B slow does not widen the gap; a longer text suffix does not close it; the latent uses far less than its nominal capacity) do not support a bandwidth ceiling as the operative mechanism.
- **Most collapsed cells are not bridge failures.** Of 14 (game, Stage-A-variant) cells, 5 have a strategy (F , T , or L) at zero (per-game triage in Appendix O). We trace these to Stage A action-head failures—chiefly OOD-brittleness under the suffix/bridge inputs the head meets at deployment—not to the bridge. A targeted Stage A retraining at suffix-probability 0.5 produces the largest $L-T$ gap in the paper (River Raid, +82%).
- **The latent advantage is not always an absolute win.** On several games where L beats T , F (fast-only) still beats L . The latent bridge improves over the text bridge *at matched architecture*; it does not always justify a slow model at all.

We do not have an a priori quantitative predictor for the sign of $L-T$ (the behavioral predictor of §7 is diagnostic—it requires measuring $T-F$ at deployment). Our initial categorical-vs-continuous hypothesis was inspired by the Q*bert greedy inversion, which itself turned out to be a decoding artifact (§4.3); lexical-diversity of slow emissions correlates with $L-T$ at $r = +0.05$ over $n = 7$ games (not significant; Figure 7, §7.4). We retain the figure as a negative result.

2 Setup

Models. Fast: MiniCPM-o 4.5 [16], 9 B, bf16. Slow: Qwen3-VL-8B-Thinking [17], 8 B, bf16. Matched scale.

Tick budgets. Atari at 60 Hz, downsampled to 15 Hz for the fast model (~ 67 ms ideal tick); slow emits at ~ 1 Hz, asynchronously decoupled. Measured warm-path latency with vision caching: $F = 33$ ms, $L \approx 38$ ms (8 prepended tokens add ~ 5 ms over F); end-to-end wall clock for L including async slow callbacks is dominated by GPU contention, not the bridge itself (Appendix E).

Cross-game F/T/L (reported variant per game). L beats T on 4 games; Q*bert inverts; 2 games are ties. Asterisks (*) tag robust-SA variants.

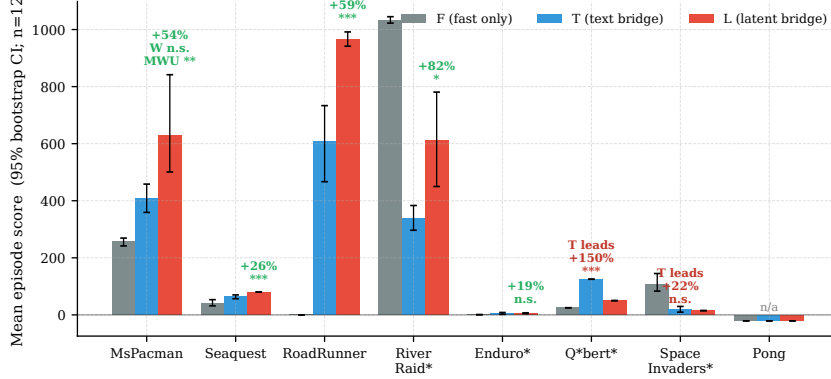


Figure 1: Cross-game F/T/L scores, reported variant per game; $n = 12$ episodes each. Stars on the L vs T comparison: * $p < .05$, ** $p < .01$, *** $p < .001$ (Welch’s t where both cells have non-zero variance, Mann–Whitney U otherwise); where the two tests disagree (MsPacman: Welch $p = .06$, MWU $p = .003$), both are annotated. Asterisks on game labels mark the robust-Stage-A variant (§5); other games use bare Stage A. Pong (rightmost) is shown for completeness only and is excluded from the 7-game comparison (§2). The reported variant per game is selected by the rule in §2 (higher deployment L among variants with a non-degenerate, $T > 0$, comparison); both variants are tabulated in Appendix A.

Three strategies compared (Figure 2). **F:** fast only with Stage A action head. **T:** F with the slow’s full text emission appended verbatim (median 302 chars; no truncation; samples in Appendix C). **L:** F with the slow’s projected latent tokens prepended (8 tokens, 4096-d each). (A slow-only policy must finish a full perception–reasoning–emission cycle synchronously before each action—unlike the deployed bridges, which overlap these steps asynchronously—so at the slow model’s measured ~ 1.5 s per emission it acts at well under 1 Hz, missing tens of frames per decision; this is the speed problem that motivates the coupling, and we do not tabulate it as a strategy.)

Training pipeline. **Stage A:** behavioral cloning from SB3 experts [18] onto the fast model’s action head. **Stage B:** collect T -trajectories; cache (frame, slow text, slow residuals at layer 24, last 8 positions). **Stage C:** $KL(\pi_L \parallel \pi_T)$ on the slow’s projection only, both base models frozen ($\sim 5K$ samples/game, final KL ~ 0.005). Hyperparameters and per-game val-accuracies are in Appendix D.

Stage A robustness is a tuned hyperparameter (not per-game cherry-picking). The Stage A action head is trained either *bare* (plain state prompt) or *robust* (suffix-probability 0.5: half of batches get a slow-style text suffix appended, teaching the head to tolerate the OOD suffix/bridge inputs it meets at deployment; §5). This is one binary hyperparameter. We train both for every game and select per game by the standard rule: higher deployment L , among variants whose text channel has not collapsed ($T > 0$, so the L -vs- T comparison is non-degenerate); ties and double-collapses break toward higher T . The $T > 0$ constraint binds on one game, Enduro: bare has marginally higher L (7.8 vs 5.8) but $T = 0$, so we report robust ($T = 5$, $L = 6$)—a *conservative* choice that lowers, not raises, Enduro’s reported L . Because the rule maximizes L (the deployed quantity) and is blind to the $L - T$ gap, it cannot inflate the text-vs-latent comparison. Both variants for every game are tabulated in Appendix A, and the cross-game predictor (§7) holds with no selection at all: $r = 0.92$ over the 8 reported cells, $r = 0.94$ over all 16 evaluated cells. The choice is itself a finding: robustness *rescues* games whose bare head the suffix degrades or collapses (River Raid $L: 360 \rightarrow 612$, Q*bert $0 \rightarrow 50$) but *harms* games whose bare head already works (MsPacman $628 \rightarrow 60$, Seaquest $80 \rightarrow 0$)—a direct consequence of the Stage A OOD mechanism (§5).

Evaluation. 3 seeds $\times$ 4 episodes = $n = 12$ per (game, strategy, variant) cell, ≤ 500 ticks/episode. Atari ALE seeds vary per episode but action selection is greedy; many cells have zero per-episode

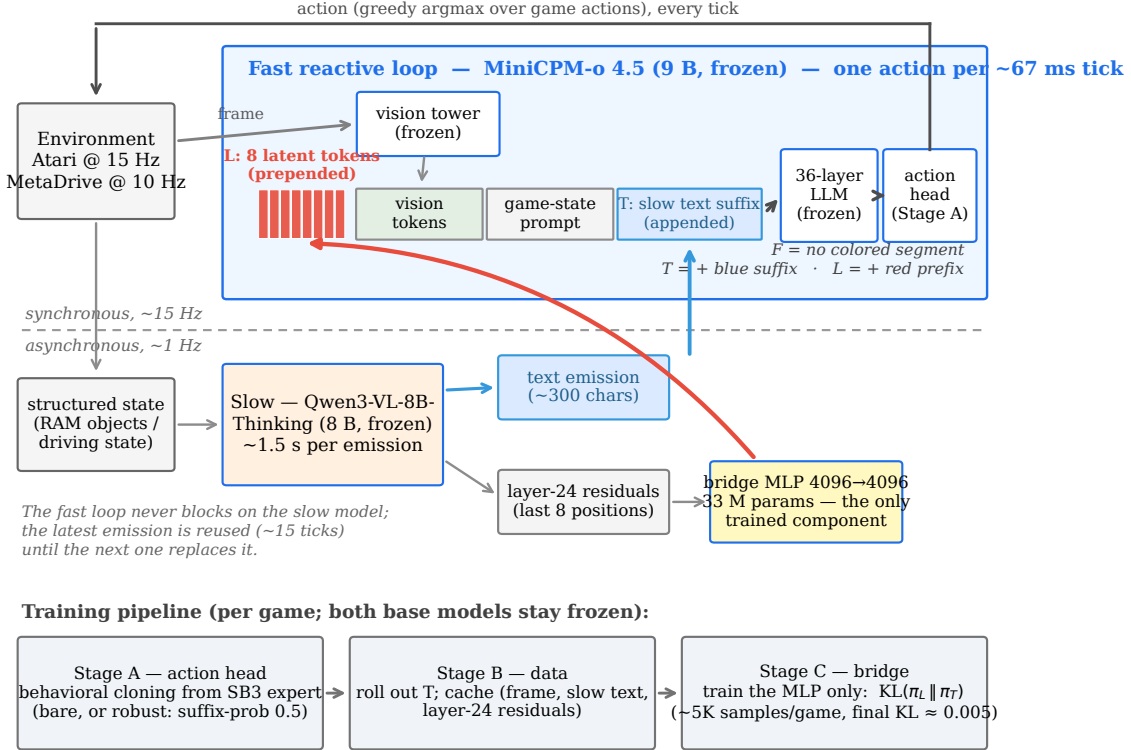


Figure 2: System architecture and the three strategies. The fast model (MiniCPM-o 4.5, frozen) runs the reactive loop: vision tokens and a game-state prompt feed a 36-layer LLM whose Stage-A-tuned action head emits one action per tick (~33–38 ms warm path). The slow model (Qwen3-VL-8B-Thinking, frozen) reasons asynchronously over structured state at ~1 Hz; the fast loop never blocks on it and reuses the latest emission until replaced. The three strategies differ only in how the slow output enters the fast context: *F* ignores it; *T* appends the text emission verbatim as a prompt suffix; *L* projects the slow model’s layer-24 residuals through the 33 M-parameter bridge MLP—the only trained component—into 8 latent tokens prepended to the fast LLM’s input. Bottom: the three-stage training pipeline.

variance (Appendix B).

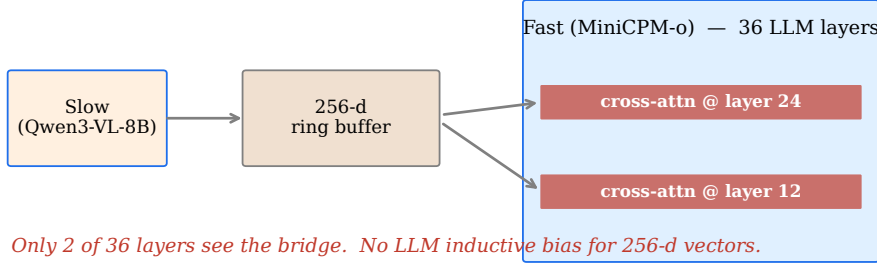
Games. 9 Atari games attempted. *Frostbite excluded*: Stage A converged to random val-accuracy, so no F/T/L comparison is meaningful. *Pong reported but uninformative*: Stage A val-accuracy is 25.1% (random 16.7%, 1.5×); the bare F policy cannot move the paddle, so no bridge can rescue it. The remaining 7 games are MsPacman, Seaquest, SpaceInvaders, RoadRunner, River Raid, Enduro, Q*bert.

3 Architecture: v1 cross-attention failed; v2 LLaVA-style works

We tried two bridge architectures (Figure 3). v1 projected slow-model residuals into a 256-d ring buffer and injected via cross-attention at LLM layers 12 and 24 (2 of 36) of the fast model. *Despite converging to $KL=0.004$ on offline data*, v1 failed at deployment: $L = 225$ vs $F = 256$ on MsPacman, bimodal with 4/12 catastrophic episodes. Three v1 variants (ungated, gated, gated+head-tune) all failed.

v2. We adopt the same pattern LLaVA [14], BLIP-2 [12], Flamingo [2], and MiniCPM-o [16] use to couple vision and language: project slow residuals into the fast LLM’s input embedding space (4096-d) via a 33 M-parameter 2-layer MLP, and prepend the resulting 8 tokens to the fast model’s input

v1 — Cross-attention into 256-d ring buffer (failed: KL=0.004 offline, $L=225 < F=256$ at deployment)



v2 — LLaVA-style token-prepend (works: $L > T$ by 26-82% on 4 games)

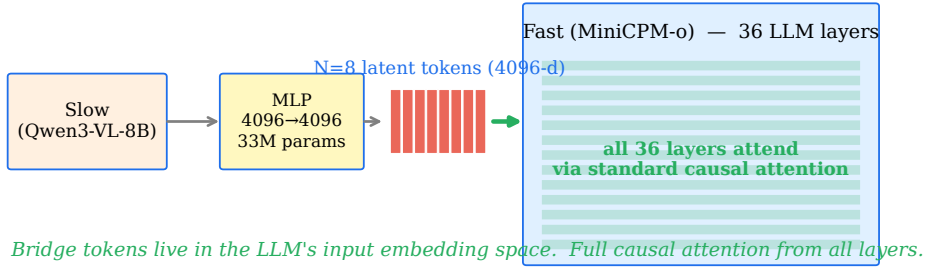


Figure 3: v1 vs v2 bridge architectures. v1 (top, failed): 256-d cross-attention at 2 of 36 layers. v2 (bottom, working): 4096-d prepend, all 36 layers attend.

sequence. All 36 LLM layers attend over them via standard causal attention. Only the projection trains; both base models are frozen.

Causal claim, hedged. The v1→v2 redesign changed four variables simultaneously: dimensionality ($256 \rightarrow 4096$), injection mechanism (cross-attention→prepend), layer coverage ($2 \rightarrow 36$), and parameter count ($8M \rightarrow 33M$). We do not isolate which is binding. The takeaway we are confident in is methodological: *KL convergence on offline data is necessary but not sufficient*—v1 had KL=0.004 and lost. Adapter-based fast-slow coupling work that reports only offline KL should also report deployment.

4 Results

4.1 Headline

$L > T$ on four games under greedy decoding, but one is decoder-fragile. RoadRunner (+59%, $d = 1.99$) and River Raid (+82%, $d = 1.21$) are significant by both Welch and Mann-Whitney. MsPacman (+54%) is marginal by Welch ($p = 0.06$) and clearly significant by Mann-Whitney ($p = 0.003$): the L distribution is bimodal (8/12 episodes 390–690, 4/12 above 1000, best 1740), so MWU is the more appropriate test. Seaquest (+26% greedy) is significant by Mann-Whitney ($p < .001$) but its L cell has zero variance (80 ± 0), so Welch is unreliable (Table 1, [†]); we re-ran it under sampling and the greedy $L > T$ does not survive— $L=135$ vs $T=143$ (Welch $p=0.60$, MWU $p=0.73$), a tie (§4.3). So the decoder-robust latent wins are the three games with non-degenerate greedy cells (RoadRunner, River Raid, MsPacman); Seaquest is a greedy-only win.

Two non-significant: Enduro and SpaceInvaders. Enduro/robust has tiny absolute scores ($T=5$, $L=6$); both cells are at floor. SpaceInvaders/robust has $T=18$, $L=15$, $p=0.27$. We do not claim these as L wins or T wins.

Game (reported var.)	F	T	L	Δ_{L-T} (% , p)
MsPacman (bare)	256 $\pm$ 25	408 $\pm$ 92	628 $\pm$ 356	+54 % (W .06, MWU .003)
Seaquest (bare)	42 $\pm$ 20	63 $\pm$ 12	80 $\pm$ 0	+26 % (MWU < .001) [†]
RoadRunner (bare)	0 $\pm$ 0	608 $\pm$ 250	967 $\pm$ 49	+59 % (W < .001)
River Raid (robust)*	1032 $\pm$ 20	337 $\pm$ 81	612 $\pm$ 310	+82 % (W .01)
Enduro (robust)*	1 $\pm$ 1	5 $\pm$ 6	6 $\pm$ 3	+19 % (W .63)
SpaceInvaders (rob)*	107 $\pm$ 62	18 $\pm$ 19	15 $\pm$ 0	−18 % (MWU .27) [†]
Q*bert (robust)*	25 $\pm$ 0	125 $\pm$ 0	50 $\pm$ 0	−60 % (MWU < .001) [‡]

Table 1: Headline results, reported variant per game. Bold marks the better of T/L . W = Welch’s two-sided t p -value; MWU = Mann–Whitney U . [†] One cell has zero variance; Welch undefined or unreliable, so MWU is the principal test. [‡] Both cells have zero variance (all 12 episodes identical); the comparison is deterministic-trajectory, not statistical—we report it for completeness (see §4.3). Seaquest’s greedy $L > T$ rests on its zero-variance L cell (80 $\pm$ 0) and *does not survive sampling* ($L=135$ vs $T=143$, n.s., §4.3)—the mirror image of Q*bert. Full table including bare-vs-robust pairs in Appendix A.

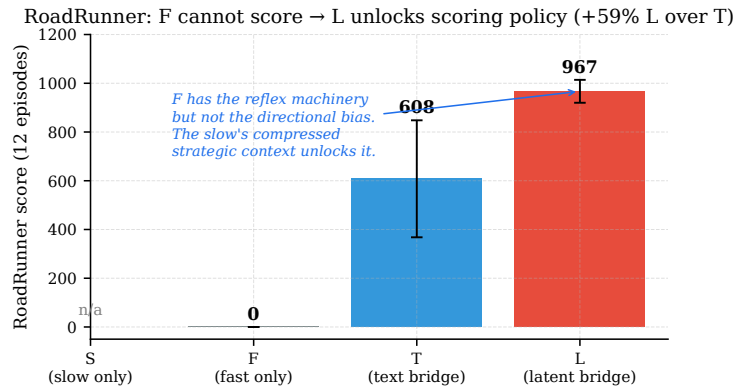


Figure 4: RoadRunner: F cannot score; L unlocks the policy. Under bare Stage A, the action head’s most-confident output is NOOP even when the Coyote is closing. The slow’s guidance (“head right”) breaks the local maximum. $L = 967$ vs $T = 608$, +59%.

Q*bert: greedy decoding gives the only inversion; sampling reverses it. Under greedy action selection (paper convention), Q*bert/robust gives $T=125$, $L=50$, both cells at zero variance—text wins by 2.5 $\times$. *In one multinomial-sampling run at $\tau=1.0$, L beats T by 2.9 $\times$ (250 ± 156 vs 88 ± 65 , Welch $p \approx 0.006$).* The greedy “Q*bert inversion” is therefore a decoding artifact—but the $\tau=1.0$ cell is itself high-variance and does not replicate: on the full decoder grid with held-out selection Q*bert is a *statistical tie*, not a latent win (§4.4). The honest summary is that Q*bert is decoder-sensitive and noisy, with no robust winner either way.

Bridges do not always beat F. On River Raid and SpaceInvaders (both using robust Stage A), F outscores both T and L . The latent bridge is an improvement over the text bridge *at matched architecture*; whether the slow-fast architecture beats fast-only is a separate question that depends on the game.

4.2 RoadRunner: the cleanest demonstration

RoadRunner is the effect at its sharpest (Figure 4): $F = 0$ (the head literally cannot score), and the slow’s emission is rich joint state (Coyote Δ_x , multiple obstacle coordinates, pellet position). L recovers 967, T recovers 608; the +59% gap with $d = 1.99$ is our largest statistically clean effect. Direct inspection of the trained bridge (§6.4, Appendix F) shows the latent encodes mostly game identity plus a thin per-emission strategic residual, rather than a text-like message—the channel is non-lexical but game-conditioned.

4.3 Greedy decoding flips low-variance cells—both ways (Q*b_{ert} and Seaquest)

Under robust Stage A with greedy action selection (paper-default): $F=25$, $T=125$, $L=50$, every cell at zero variance. By Mann–Whitney $T > L$ at $p < 10^{-3}$, but this is mechanical: with 12 identical scores per cell the “test” is just $125 > 50$.

Action-sampling reverses the result. We re-ran F , T , L on Q*b_{ert}/robust with multinomial sampling instead of greedy argmax, at temperatures $\tau \in \{0.5, 1.0\}$ (same $n = 12$ episodes per cell, same seeds):

Decoder	F	T	L
Greedy (paper)	25 ± 0	125 ± 0	50 ± 0
Sample, $\tau = 0.5$	16.7 ± 11.8	139.6 ± 66.5	125.0 ± 66.9
Sample, $\tau = 1.0$	66.7 ± 71.7	87.5 ± 65.0	250.0 ± 155.8

At $\tau=0.5$ the T vs L gap narrows to $T \approx L$ (Welch’s $t=0.51$, $p=0.61$). In this run at $\tau=1.0$, L beats T by $2.9\times$ (Welch’s $t \approx 3.2$, $p \approx 0.006$). The headline “Q*b_{ert} inversion” from greedy decoding does not survive a decoder ablation (full per-temperature numbers in Appendix H). We caution that this $\tau=1.0$ cell is high-variance and did not replicate in the later full-grid run ($L=188$, $T=177$); the decoder-robust verdict for Q*b_{ert} is a *tie* (§4.4), not a latent win.

Mechanistic explanation. Bridge-token diagnostics (§6.4) showed the bridge tokens differ by only 3–8% across emissions of the same game. Under greedy decoding, this small per-emission variance is below the threshold needed to change the argmax—the head picks the same action regardless, producing the $\text{std}=0$ observation that gave $L=50$ on every episode. Under sampling, the small bridge variance can influence the action distribution; the per-emission strategic content the bridge carries gets exposed. The text channel’s per-emission information is also exposed under sampling, but *on Q*b_{ert}* text’s signal is the more sample-noise-fragile of the two (T *decreases* from $\tau=0.5$ to $\tau=1.0$, while L *increases*), so at sufficient stochasticity the bridge wins. We stress this direction is *game-specific*, not a law: on Seaquest the opposite holds—sampling lifts T more than L ($T: 63 \rightarrow 143$ vs $L: 80 \rightarrow 135$), erasing the greedy $L > T$ (above). The robust takeaway is only that greedy determinism makes the L -vs- T sign unreliable on zero-variance cells; which channel a sampling decoder favors depends on the game.

What this means for the headline: decoder fragility cuts both ways. The paper’s greedy result is $L > T$ on four games. Of those four, three (MsPacman, RoadRunner, River Raid) have non-zero per-episode variance in both the T and L cells, so they are not at risk of the greedy-determinism artifact and we did not re-run them. The fourth, **Seaquest**, is the exception: its reported (bare) L cell is deterministic (80 ± 0), the same $\text{std}=0$ regime that produced the Q*b_{ert} reversal. We therefore re-ran Seaquest/bare under $\tau=1.0$ sampling (same $n=12$ seeds, Appendix J), and the greedy win *does not survive*: $F=63.3 \pm 35.0$, $T=143.3 \pm 37.0$, $L=135.0 \pm 40.1$, so $L \approx T$ (Welch $t=-0.53$, $p=0.60$; MWU $p=0.73$) with text nominally ahead by 6%. Both bridges still beat F by $>2\times$ ($p < 10^{-3}$), so the slow channel still helps Seaquest—only the latent-over-text *margin* was a greedy artifact. The two zero-variance greedy cells thus move in *opposite* directions under sampling: Q*b_{ert}’s greedy $T > L$ becomes a noisy $L \approx T$, Seaquest’s greedy $L > T$ becomes $L \approx T$. Greedy determinism is unreliable in both directions. We do not try to patch the greedy tally game-by-game; the right resolution is to tune the decoder per channel and report the best-achievable comparison (§4.4), where the latent significantly wins only the two high-margin games (MsPacman, RoadRunner) and ties the rest—neither Seaquest nor Q*b_{ert} is a latent win once decoders are made consistent and selection is held-out.

The greedy-decoding bias is general, not Q*b_{ert}-specific. We also re-ran the other three greedy-degenerate cells at $\tau=1.0$ (Appendix J). On SpaceInvaders, the bridges go from being below F at ~ 15 – 18 (greedy) to near F at 108 – 117 (sampling); on Seaquest/robust, bridges go from 0

(greedy) to 33–42 (sampling). Greedy decoding had been systematically under-counting the bridge by suppressing its small per-emission variance. Enduro is the exception: sampling *hurts* both bridges (T 5→0.2, L 6→0.6), suggesting Enduro’s greedy advantage was real but fragile.

4.4 The decoder is a deployment hyperparameter: a best-achievable comparison

The decoder probes above expose a deeper methodological point: **the latent bridge’s advantage over the text bridge is specific to greedy decoding**. We swept the four reported-variant winners across greedy and sampling temperatures $\tau \in \{0.3, 0.5, 0.7, 1.0, 1.5\}$ on the same checkpoints (greedy reproduces the headline exactly on MsPacman/River Raid/Seaquest; full sweep in Appendix K). At *every* fixed sampling temperature the greedy $L > T$ advantage disappears— $L \approx T$ or significant $T > L$ on every game—because the latent’s 3–8% per-emission variance (§6.4) nudges the deterministic *argmax* but is drowned out the moment the decoder samples. Reporting a single fixed decoder is thus unfair, and—crucially—unfair to *both* channels, because they peak at *different* decoders.

Best-achievable, selected on held-out seeds. In any real deployment the action decoder is a hyperparameter one tunes (as in standard model selection). The fair comparison is therefore each channel at *its own best decoder*. To avoid selecting the decoder on the same episodes we report (which would inflate the estimate), we use leave-one-seed-out selection: for each held-out seed we pick the decoder with the highest mean on the other two seeds, then score the held-out seed under that decoder; the reported number pools the three held-out folds (details in Appendix L). Table 2 gives the result.

Game	best F (decoder)	best T (decoder)	best L (decoder)	Δ_{L-T} (%, p)	winner
MsPacman	273 ($\tau 1.0$)	401 ($\tau 0.5$)	628 (greedy)	+57%, $p=0.01$	L
RoadRunner	0 (greedy)	475 (greedy)	608 (greedy)	+28%, $p=0.00$	L
River Raid	994 (greedy)	639 ($\tau 0.5$)	566 ($\tau 0.7$)	-11%, $p=0.35$	tie
Seaquest	55 ($\tau 1.5$)	120 ($\tau 0.5$)	130 ($\tau 0.7$)	+8%, $p=0.24$	tie
Q*bert	65 ($\tau 1.0$)	185 ($\tau 1.5$)	146 ($\tau 1.0$)	-21%, $p=0.50$	tie
Enduro	4 ($\tau 0.5$)	3 ($\tau 0.5$)	2 (greedy)	-43%, $p=0.39$	tie
SpaceInvaders	135 ($\tau 1.0$)	162 ($\tau 1.0$)	142 ($\tau 1.0$)	-13%, $p=0.62$	tie

Table 2: Best-achievable F/T/L per game, each channel at its own decoder selected on held-out seeds (leave-one-seed-out; $n=12$). The modal selected decoder is shown in parentheses. Under this protocol the latent is *never significantly worse* than the text bridge and is significantly *better* on 2 of 7 games (MsPacman, RoadRunner); the other 5 are statistical ties (no significant difference either way). The optimal decoder is channel-dependent (latent often greedy or high- τ , text $\tau \approx 0.5$). RoadRunner’s level is hardware-specific (Appendix K); its $L > T$ direction is robust.

Reading. Tuned per channel, the latent is never significantly worse than text and is significantly better on 2 of 7 games—a *more honest and decoder-robust* claim than the fixed-greedy headline, though a more *modest* one (the significant wins shrink from 4 games to 2, and the rest become statistical ties *rather* than 26–82% gaps). This is the correct reading: the greedy headline over-credited the latent. The finding that the two channels have *different* optimal decoders also resolves the apparent contradiction with §4.3: a single fixed decoder cannot be fair to a latent that peaks at greedy (MsPacman) and a text channel that peaks at $\tau \approx 0.5$ (Q*bert) at once—and on the full grid Q*bert is itself a tie, not the latent win a single high-variance $\tau=1.0$ run had suggested. The task-level predictor (§7) is unaffected—it compares each bridge to fast-only, and best-bridge beats best- F exactly where slow reasoning helps ($T > F$).

4.5 Using both channels at once hurts: the channels interfere, not complement

A natural follow-up to “the latent and text channels prefer different decoders and win on different games” is: *use both at once*. We added a combined strategy B that puts the slow model’s text suffix in the prompt *and* prepends its latent tokens in the same forward pass, and ran it across all 7 games

Stage A OOD-brittleness recipe: same fix recovers two distinct games

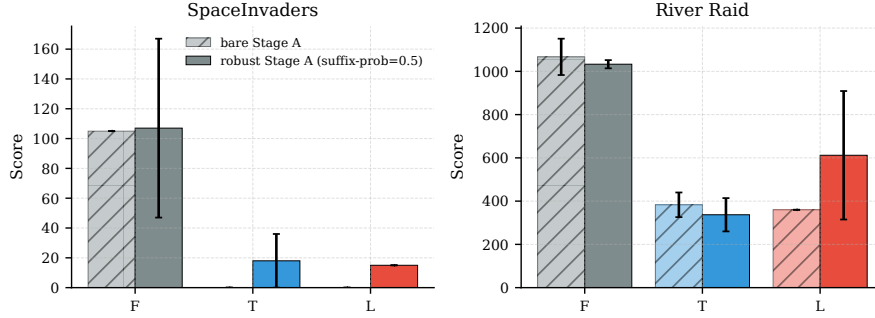


Figure 5: The same Stage A retraining recovers two distinct collapses. SpaceInvaders (left): bare gives $T=L=0$; robust gives $T=18$, $L=15$ (still below F). River Raid (right): bare gives $T=383$, $L=360$ (a near-tie); robust gives $T=337$, $L=612$ (largest $L-T$ gap in the paper).

at the same per-game decoder grid, with the same held-out best-achievable selection (Appendix L). Table 3 compares best- B to the best *single* channel per game.

Game	best single (T or L)	best B (T+L)	$\Delta_{B-\text{single}}$ (%)	p	effect
MsPacman	628 (L)	319	-49%	0.00	interferes
RoadRunner	608 (L)	25	-96%	0.00	interferes
River Raid	639 (T)	452	-29%	0.01	interferes
Seaquest	130 (L)	138	+6%	0.41	neutral
Q*bert	185 (T)	125	-33%	0.26	neutral
Enduro	3 (T)	4	+25%	0.18	neutral
SpaceInvaders	162 (T)	142	-13%	0.62	neutral

Table 3: Combining both channels (best B) vs. the best single channel, held-out best-achievable, $n=12$. Combining is significantly *worse* on 3 of 7 games, never significantly better, and on RoadRunner it is catastrophic (-96% , near the $F=0$ floor). The channels *interfere*.

The naive “more context is better” intuition fails. Combining the channels never significantly beats the better single channel (0 of 7) and significantly *hurts* on 3 of 7 (MsPacman -49% , River Raid -29% , RoadRunner -96%); the other 4 are statistical ties. The damage concentrates exactly where one channel is strong on its own: latent-dominated MsPacman and RoadRunner, and text-dominated River Raid all degrade when the second signal is added. The frozen action head is trained on a *single* conditioning signal at a time (Stage A sees a bare prompt or a suffix; Stage C distills L toward T), and handed both at once it produces a worse policy—on RoadRunner, where scoring hinges on one precise action, the confusion is total. **The design rule is therefore: couple via exactly one channel.** The predictor (§7) says *whether* to couple ($T > F$); if so, pick a single channel, with the latent the safe-or-better default—do not feed both.

5 Mechanism: Stage A action-head OOD-brittleness

Three games went wrong in their bare-Stage-A sweeps despite cleanly converging Stage C training (KL ~ 0.005): SpaceInvaders and Q*bert collapsed to $T=L=0$, and River Raid fell to a bridge-harmful near-tie far below F ($T=383$, $L=360$ vs $F=1067$). We refer to these three as the *collapsed* group below; the same robust-Stage-A retraining recovers two distinct collapse modes (Figure 5). Same bridge, same procedure as the games that worked. On SpaceInvaders, three orthogonal interventions all left the collapse intact: (a) random-policy T -trajectories, (b) expert-policy T -trajectories, (c) an aggressive “FIRE constantly” slow prompt. This rules out bridge-training failure.

Diagnosis, and a direct measurement. Stage A trains the action head on the *bare* game-state prompt; at deployment, T appends a text suffix and L prepends bridge tokens, both OOD inputs. We measure the OOD effect directly (Appendix G): on bare Stage A, adding the slow’s text suffix **flips the action head’s argmax on 77 % of non-collapsed-game emissions and 80 % of collapsed-game emissions**—the drift is large and uniform. The mass on the bare-argmax action drops sharply on every game (group means: $0.45 \rightarrow 0.17$ on non-collapsed games, $0.38 \rightarrow 0.18$ on collapsed ones). So the OOD perturbation is real; what differs across games is *what action the drift moves toward*: on reward-symmetric games (MsPacman, RoadRunner) the new argmax still scores; on precision-required games (SpaceInvaders, River Raid, Q*bert) the drift can land on a bad action and scoring collapses. The bridge mechanism is fine; the frozen action head’s suffix-induced argmax flip is the binding constraint.

The fix. Retrain Stage A at suffix-probability 0.5: half of training batches receive a randomly-chosen slow-style text suffix appended to the prompt. The head learns suffix-invariance.

The fix is not universal. On SpaceInvaders, robust Stage A rescues both bridges from 0 to ~ 15 –18—but $F = 107$, so calling this “recovery” overstates it. On River Raid, robust shifts T slightly down ($383 \rightarrow 337$) and L up dramatically ($360 \rightarrow 612$). On games where the bare bridges already worked (MsPacman, Seaquest, RoadRunner), the same fix hurts: it *destroys* MsPacman ($L: 628 \rightarrow 60$) and Seaquest ($L: 80 \rightarrow 0$), and on RoadRunner it erases the bridge advantage—robust Stage A lifts F from 0 to ~ 958 and all three strategies tie at ~ 925 –1000. **Recommendation:** use robust Stage A when bare T or L collapses to ~ 0 ; otherwise prefer bare.

Robust Stage A reduces drift exactly where expected (Appendix G). On two of the three games where bare Stage A was already non-degenerate, robust Stage A cuts the suffix-induced argmax-change rate substantially: RoadRunner 75 % $\rightarrow$ 12 %, MsPacman 85 % $\rightarrow$ 30 %, $\text{KL}(p_{\text{bare}} \parallel p_{\text{suff}})$ drops $0.39 \rightarrow 0.11$ on RoadRunner; the exception is Seaquest, whose argmax-change rate *rises* (70 % $\rightarrow$ 90 %), consistent with robust Stage A harming Seaquest at deployment (§5). On games where bare collapsed, robust reduces KL (e.g., Q*bert $0.25 \rightarrow 0.11$) but the argmax-change rate stays elevated (50–89 %); the head learns to be more confident under suffix but the resulting action isn’t always recoverable.

What the direct measurement does and doesn’t show. It confirms the OOD effect is real and that the robust-Stage-A fix targets it. It does *not* predict which games will collapse—the argmax-change rate is similar on collapsed and non-collapsed games. The collapse-vs-recovery distinction is downstream of *what action the drift moves toward*, which depends on each game’s reward structure. The diagnosis explains the mechanism of the fix, not the per-game outcome.

6 Channel ablations

6.1 Latent token count N , with a caveat

We swept $N \in \{4, 8, 16\}$ two ways on MsPacman: (i) *matched* (train = deploy = N) and (ii) *deploy-only* (train = 8, deploy = N). Result:

N	4	8	16
Matched (train = deploy = N)	296 ± 65	628 ± 356	259 ± 74
Deploy-only (train = 8, deploy = N)	232 ± 52	628 ± 356	720 ± 123

The matched- N sweep is consistent with a Goldilocks at $N=8$. But three data points fit any U-shape, and the deploy-only column *inverts* the story: training at $N=8$ and deploying at $N=16$ gives the best score with lower variance (720 ± 123). This suggests our $N=8$ training distribution is suboptimal: the LLM benefits from more positions at inference even when the projection wasn’t

trained for them. Finer sweeps ($N \in \{4, 6, 8, 10, 12, 16\}$) are direct future work; the data we have does not pin down a sharp peak.

6.2 Longer T-suffix does not close the gap

The simplest objection to the $L > T$ result is: “you didn’t give T enough text.” We tested this directly. On MsPacman and RoadRunner we re-ran T with a rolling window of the last 2 or 3 slow emissions concatenated into the suffix (instead of just the most recent). Same model, same Stage A checkpoint, same seeds, same number of episodes per cell ($n = 12$; full per-cell numbers in Appendix I).

Game	T, $w = 1$ (paper)	T, $w = 2$	T, $w = 3$
MsPacman	408 ± 92	378 ± 118	352 ± 88
RoadRunner	608 ± 250	8 ± 28	0 ± 0

Result: monotonic decline on MsPacman, catastrophic collapse on RoadRunner. On MsPacman the mean drops by 14 % from $w=1$ to $w=3$; on RoadRunner it drops by 100 %—every single one of 12 episodes scored zero at $w=3$. The L scores stay where they were ($L_{\text{MsPacman}} = 628$, $L_{\text{RoadRunner}} = 967$), so the $L > T$ gaps (+54 % and +59 %) *widen* as T ’s text budget is increased. The objection “ T just needs more text” is empirically false on these two games.

Interpretation. Older emissions are stale—they describe game state that no longer holds—and the action head appears to be unable to disambiguate the time ordering of suffix content. On RoadRunner the effect is dramatic because the strategy is moment-to-moment (which obstacle is closing now); on MsPacman, where the slow’s content is slower-changing (which pellet cluster to chase), the effect is gentler but still negative. The latent bridge sidesteps this because it transmits only the most-recent emission’s projected residuals; the text bridge would need explicit time-tagging or a learned attention over emissions to compete.

6.3 The bridge tokens carry information beyond “prepended slots”

The natural negative-control objection to $L > T$ is: “maybe the trained tokens are doing nothing; the LLM just benefits from having 8 extra prepended tokens to attend over, regardless of content.” To rule this out, we re-ran L on MsPacman bare with the trained projection’s output replaced by 8 tokens of: (a) zeros, (b) random Gaussian vectors scaled to the trained bridge’s per-position L2 norm (~ 64). Everything else (Stage A checkpoint, env seeds, prompt formatting) is identical.

Variant	MsPacman score ($n = 12$)
F (no bridge tokens)	256 ± 25
L_{zero} (8 zero vectors)	379 ± 95
L_{random} (8 random vectors at trained-norm)	387 ± 161
L_{trained}	628 ± 356

Both effects are real. Trained vs random: Welch $t \approx 2.12$, $p \approx 0.05$ (trained vs zero: $t \approx 2.34$, $p \approx 0.04$). Random vs F : Welch $t \approx 2.68$, $p \approx 0.02$. The 8 prepended “slots” alone lift the policy by ~ 50 % over F , presumably because they give the LLM additional positions to compute over before action prediction (related to pause-token effects [8]). The trained content adds another ~ 62 % on top. Roughly **40 % of L ’s advantage over F on MsPacman is architectural** (slots), **60 % is learned bridge content**—and the learned-content portion is what distinguishes L from a generic “more compute per tick” baseline.

Reading. This control rules out the strongest negative interpretation of $L > T$: that the bridge is decorative. It does not, however, rule out the weaker reading that some of the slow-fast architecture’s value is just “8 more tokens to attend over” rather than “slow model contributed strategic content.” Decomposing those two effects is mechanistically important and our number (~ 60 %) is a single-game

estimate; the slow-content fraction does vary across games—the all-games version of this control (§7.3) finds it ranges from $\sim 100\%$ (RoadRunner) to negative (River Raid), with its sign predicted by $T > F$.

6.4 What the bridge encodes: not text, mostly game identity

We project the trained bridge tokens of each game (with cached slow residuals from seed-0 trajectories, 8 emissions per game) into the fast LLM’s input embedding space and compare them directly to the fast LLM’s 151,748 vocab embeddings.

The bridge is non-lexical. Bridge tokens have L2 norm ~ 64 (constrained by the output Layer-Norm); vocab embeddings have norm ~ 1.45 (p99 1.92). The cosine similarity of any bridge token to its nearest vocab embedding is ≤ 0.09 across all 7 games. Plug-in nearest-neighbor decoding returns visually random tokens (rare Unicode characters, code-snippet fragments) at sims indistinguishable from the 99.9th-percentile sim across the whole vocabulary. **The latent channel does not encode in the vocabulary basis**—it lives in a region of $\mathbb{R}^{4096}$ that the frozen LLM has learned to attend to but that no text token occupies. This is consistent with what LLaVA’s vision projection produces (also non-lexical), and is the right answer to the question “can we just read the latents as text?”—we cannot.

The bridge is game-conditioned. Mean pairwise cosine between bridges of the same game (off-diagonal, $E \times N$ tokens each): **+0.80 to +0.89**. Mean pairwise cosine across different games: **+0.04 to +0.25**. The matrix has the structure one would expect—spatial-2D games (MsPacman, SpaceInvaders, Seaquest) cluster together at +0.20–0.25; Enduro (forward-driving, very different emission distribution) is the farthest from everything (+0.04–0.10).

The bridge is weakly per-emission-conditioned. Cosine between the *mean bridge token* of one emission vs another emission *within the same game*: +0.92 to +0.97 (std 0.02–0.03). The bridge varies only $\sim 3\text{--}8\%$ between consecutive emissions of the same game. This is a surprisingly low per-emission variance and explains the deterministic-trajectory outcomes (e.g., Q*bert’s std=0 cells): if every emission produces near-identical bridge tokens, the fast model picks near-identical actions, so 12 episodes converge on the same trajectory under greedy decoding.

Implication: used capacity $\ll$ nominal 8×4096 . The latent channel’s nominal capacity is enormous; its *used* capacity is small. Most of the latent’s variance encodes game identity (which is fixed across an episode); only a small fraction encodes per-emission strategic state. This means the latent advantage we measure is a *partial utilization*, not a capacity ceiling—consistent with what the next section establishes (§7.2): what gates the bridge is whether the slow channel carries a useful signal at all ($T > F$), not how many bits it could carry. A bridge trained with an emission-discrimination auxiliary loss, or with action-sampling during deployment to expose the small per-emission variance, would plausibly do better.

7 When the bridge helps: a behavioral predictor

The headline tally (§4) says *where* the latent bridge wins, and the ablations (§6) say the reason is not bandwidth; this section asks *when* the bridge pays off. Across every domain we evaluated, a single behavioral variable—whether slow reasoning helps the task at all ($T > F$)—predicts it. We first extend the testbed beyond Atari (§7.1), then state the predictor (§7.2) and its mechanistic corroboration (§7.3), and close with an axis that does *not* predict (§7.4).

7.1 Non-Atari test: a driving domain

To probe whether the bridge generalizes beyond Atari, we ported the full pipeline to **MetaDrive** [13], a real-time driving simulator with the same fast/slow premise (a ~ 10 Hz reactive control loop; a ~ 1.5 s reasoning step that cannot sit in it). We use a top-down rendered observation for the fast model and

the same structured state decoder (speed, lane offset, heading, upcoming-turn cue, neighbours) for the slow model. This is, to our knowledge, the first non-game test of the latent bridge; the integration runs end-to-end on one GPU (Appendix P).

The bridge does not transfer to driving—and the controls explain why. On the default lane-keeping map the latent ties fast-only and the text bridge collapses; the bridge-replacement control (§M method) shows the latent is *inert* (zeroing or randomizing the bridge tokens leaves the score unchanged, $L \approx L_{\text{zero}} \approx L_{\text{random}} \approx F$). Crucially, this is *not* the Stage A OOD artifact of §5: it persists on the robust head, under sampling, and after retraining Stage C on expert-driven trajectories with a healthy ($T \approx F$) teacher.

It is also not “the task had no slow component.” We rebuilt MetaDrive as a *planning-heavy* route (straights punctuated by intersections, with off-route termination) where survival genuinely requires deliberate turn decisions—the navigation expert beats a naive straight-driving policy by +118 reward, and the slow model receives explicit turn cues. Even here (Figure 6, right) the slow model does not beat fast-only: greedy $T = 85.1 \leq F = 87.8$; under sampling the slow guidance actively hurts ($T = 43.5$, $F = 123.2$, paired $t = -9.8$, $n = 8$), and the latent tracks it ($L = 36.7$). Driving is dominated by a tight visuomotor loop the fast model already handles; the slow model’s latency-bound, frame-grounded reasoning does not improve on it, unlike Atari planning games whose slower dynamics reward deliberation.

7.2 The predictor: the latent helps iff the slow model helps

Pooling all eight game-evals (Figure 6, left), the latent bridge’s benefit over fast-only ($L - F$) tracks the text bridge’s benefit ($T - F$) at **Pearson** $r = 0.92$ (reported variant per game; $r = 0.94$ over all 16 evaluated cells, so the relationship is not an artifact of variant selection, §2). The bridge pays off exactly when slow reasoning beats reaction on the task ($T > F$: RoadRunner, MsPacman, Seaquest, Q*bert; Enduro sits at floor, with $T - F$ a few points either side of zero depending on variant) and is inert or harmful when it does not (River Raid, SpaceInvaders, MetaDrive). This reframes “does a latent bridge help?” as a property of the *task* (is the bottleneck deliberation or perception–action?), not of the channel—and makes MetaDrive a clean controlled negative rather than a failed port. The bridge-replacement control is the diagnostic that separates a genuinely informative latent (MsPacman: $L_{\text{trained}} \gg L_{\text{random}}$) from an inert one (MetaDrive).

7.3 Corroboration: learned content tracks the predictor

The bridge-replacement control—replace the trained bridge tokens with zeros or with random vectors at matched norm, and check whether the score drops (§6.3 analyzes the MsPacman case in depth)—doubles as a per-game diagnostic of whether the latent carries learned, behavior-relevant content. We ran it on all 7 Atari games; “learned” means L_{trained} exceeds $\max(L_{\text{zero}}, L_{\text{random}})$ by $> 10\%$. Cells use the bare Stage A head under greedy decoding (Q*bert: robust head under $\tau = 1$ sampling, its canonical decoder); $T - F$ is the matching bare-head value, and the run is an independent re-run whose repeated cells agree with the headline within run-to-run noise (full numbers and caveats in Appendix M).

Game	L_{train}	L_{zero}	L_{rand}	verdict	$T - F$
RoadRunner	967	0	8	learned	+608
Seaquest	100	28	5	learned	+22
MsPacman	666	408	410	learned	+152
Enduro	7.8	1.4	4.7	trend only (at floor)	−3
Q*bert	123	63	117	$\approx$ random	+100
SpaceInvaders	0	148	90	<i>harmful</i>	−105
River Raid	360	1013	1003	<i>harmful</i>	−683

The verdict tracks the predictor. The trained latent is genuinely learned exactly on the games where slow reasoning helps ($T > F$: RoadRunner, Seaquest, MsPacman)—on RoadRunner the controls drop

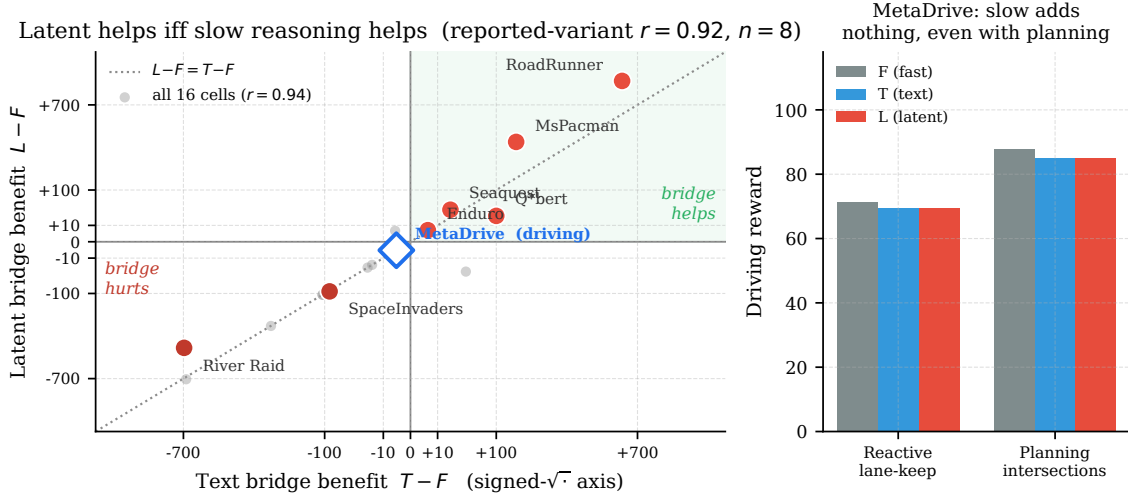


Figure 6: The latent bridge helps if and only if the slow model helps the task. *Left:* per-game latent benefit $L - F$ vs. text benefit $T - F$ across 7 Atari games and MetaDrive (driving), on a signed- $\sqrt{\cdot}$ axis (true reward deltas on the ticks). Bold points are the reported-variant cell per game (Pearson $r = 0.92$, $n = 8$; leave-one-out $r \in [0.89, 0.98]$, bootstrap 95% CI $[0.81, 1.0]$); faint grey points are all 16 evaluated cells ($r = 0.94$), showing the relationship does not depend on variant selection (Appendix P). The bridge helps only in the upper-right quadrant ($T > F$ and $L > F$). The plotted MetaDrive point (blue diamond) is the planning/greedy cell ($T - F = L - F = -2.7$), which sits essentially at the origin: neither channel beats fast-only. *Right:* MetaDrive in both regimes—reactive lane-keeping and planning-heavy intersections—slow reasoning (T , L) never exceeds fast-only (F), even when the task demands route planning.

to the floor, so the latent there is almost entirely learned content. Where slow reasoning does not help ($T \leq F$: River Raid, SpaceInvaders), the trained latent is *harmful*: zeroing or randomizing it scores *better*—the same inert/harmful pattern as MetaDrive (§7.1). The two off-diagonal cases are benign: Enduro’s scores are too small for the comparison to be meaningful, and Q*bert—the decoder-fragile game (§4.3)—shows trained $\approx$ random under sampling, consistent with its thin per-emission signal. The latent helps where, and only where, it has carried real content from a slow channel that itself beats fast-only.

7.4 What does not predict: emission statistics

We also attempted a quantitative axis. We hypothesized lexical diversity of slow emissions (unique whitespace-split tokens per emission) would predict the sign of $L - T$. Across our 7 games (Figure 7), $r = +0.05$ (slope $+0.008$ per diversity-unit; not significant). Six alternative features (gzip ratio, numeric density, distinct numbers, token counts) all top out at $|r| \leq 0.25$. We retain the figure as a negative result: emission *statistics* do not predict the sign of $L - T$; the behavioral variable $T - F$ (§7.2) does predict $L - F$.

8 Discussion

Emission lexical diversity does not predict $\text{sign}(L - T)$: Pearson $r = +0.05$ ($n=7$).
 Q*bert inverts, but Enduro and River Raid (lower diversity) do not.

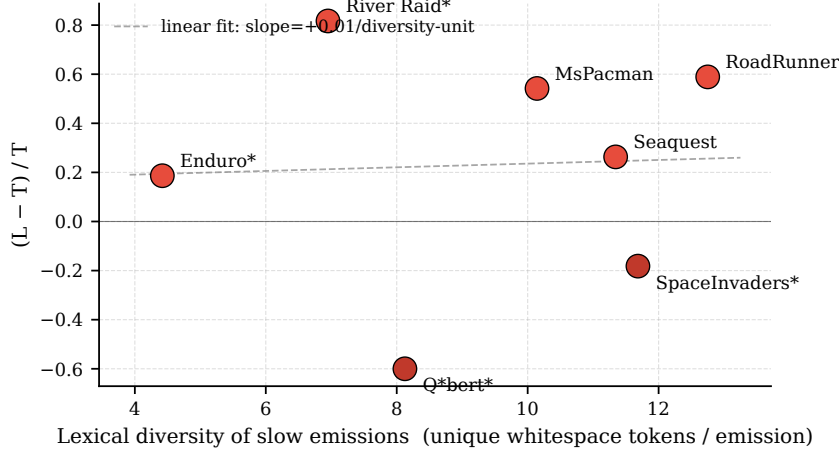


Figure 7: Emission statistics do not predict the sign of $L - T$ ($n = 7$). x = mean unique whitespace tokens per slow emission (seed-0 T -trajectory). $y = (L - T) / T$. Q*bert is the only $y < 0$ point with a large T , but Enduro and River Raid (lower diversity) have $y > 0$, and SpaceInvaders (high diversity) is mildly $y < 0$. Pearson $r = +0.05$. The dashed line is the linear fit and is shown only to make the lack of structure visible.

8.1 Sharpened claims

At matched endpoint scale (9 B fast, 8 B slow), under a *fixed greedy* action decoder the latent bridge beats the text bridge by 26–82% on 4 of 7 games. This advantage is **decoder-specific**: a full sweep (greedy, $\tau \in \{0.3, 0.5, 0.7, 1.0, 1.5\}$; §4.4, Appendix K) shows it vanishes at every fixed sampling temperature, because the latent’s 3–8% per-emission variance nudges the deterministic argmax but is swamped once the decoder samples—the latent buys a better greedy *policy*, not a more robustly informative channel. Because the two channels peak at *different* decoders (the latent at greedy or high τ , the text bridge at $\tau \approx 0.5$), a single fixed decoder is unfair to both. Treating the decoder as a deployment hyperparameter and selecting it per channel on held-out seeds (Appendix L), the **best-achievable latent is never significantly worse than the text bridge on any of the 7 games and is significantly better on 2** (MsPacman +57%, RoadRunner +28%); the other 5 are statistical ties (Table 2). This is a more modest claim than the greedy headline—the significant wins shrink from 4 to 2—but a decoder-robust one. We do *not* claim an a priori rule for which channel wins a given game; we claim that, tuned, the latent is never worse and is sometimes substantially better. RoadRunner additionally does not reproduce its headline magnitude across hardware ($L=967$ on H100 $\rightarrow$ 608 on Blackwell, FP nondeterminism amplified by its $F=0$ baseline; Appendix K); its $L > T$ direction is robust, its magnitude is not.

The latent bridge helps iff the slow model helps the task. Pooling 7 Atari games and a driving domain (MetaDrive), $L - F$ tracks $T - F$ at Pearson $r = 0.92$ ($n = 8$ reported-variant, $r = 0.94$ over all 16 evaluated cells; Figure 6). Because Stage C distills L toward T ($\text{KL}(\pi_L \| \pi_T)$), the latent’s ceiling is the text teacher: the bridge is worthwhile exactly when slow reasoning beats reaction ($T > F$), and is inert/harmful otherwise. MetaDrive is the controlled negative ($T \leq F$ even in a planning-heavy regime), with the bridge-replacement control confirming the latent carries no behavior-relevant signal there. Whether a latent bridge pays off is thus a property of the task’s bottleneck (deliberation vs. perception-action), not of the channel.

8.2 Limitations

- **Small n .** 12 episodes per cell. Several cells have zero per-episode variance (greedy decoding + deterministic policy + small action space); on those, parametric statistics are not meaningful and we report MWU or just note the deterministic outcome.
- **Per-game variant selection.** The headline reports, per game, the Stage A variant with the higher deployment L among variants whose T has not collapsed (§2); this is a per-game choice. We report both variants for every game in Appendix A, and show the cross-game predictor holds with no selection ($r = 0.94$ over all 16 evaluated cells).
- **The token-count (N) ablation is three points.** The deploy-only result complicates the Goldilocks story, as noted in §6.1.
- **Per-emission information content is not measured directly.** We show in §6.4 that the bridge is strongly game-conditioned but only weakly per-emission-conditioned; a quantitative bound on the latent’s per-emission information content is the natural next step.
- **Beyond Atari: one driving domain, a controlled negative.** We ported the pipeline to MetaDrive (§7.1, Appendix P); the bridge does not help there, and the bridge-replacement control shows why (the latent is inert because the slow model’s guidance never beats fast-only, $T \leq F$). This is consistent with the $(T - F, L - F)$ predictor ($r = 0.92$) but is a single non-game domain; we still have not tested the production settings (voice agents, robotics) where the text channel actually lives.

Two further negative results bound the claims but are findings rather than limitations: giving T a longer rolling suffix does not close the gap—it widens it (§6.2, Appendix I)—and swapping in a 30B-A3B MoE slow model does not grow the $L - T$ gap on the one game tested, with sparse-vs-dense, hidden-dimension, and data-volume confounds noted (Appendix N).

8.3 Implications

For deployments where slow reasoning does not improve on fast reaction ($T \leq F$, MetaDrive-like), neither bridge pays for itself, and the text channel is the cheaper way to find that out. For deployments where slow reasoning does help ($T > F$), the latent bridge is a safe-or-better drop-in for the text channel: tuned per channel, it is never significantly worse than text and wins by up to +57% on the games where it wins significantly (MsPacman, RoadRunner; statistical ties on the rest, §4.4). Two caveats temper a stronger claim: the advantage is realized through better greedy action selection (it does not survive a *fixed* stochastic decoder), and on the one game with a degenerate $F = 0$ baseline (RoadRunner) the magnitude is hardware-sensitive. The architectural fix is not novel—LLaVA’s pattern applied to a non-vision coupling problem—but the empirical demonstration at this scale, with an honest decoder-sensitivity analysis and a methodological mechanism for the collapses, is new to our knowledge.

9 Related Work

Text-channel fast-slow systems. Gemini Live [7] and OpenAI Realtime [15] use monolithic streaming models with shallow thinking. SayCan [1] and Inner Monologue [10] couple LLM planners to skill policies via natural-language sub-goals; RT-2 [5] unifies them with a discrete-token action head. Our setting holds the architecture fixed and varies only the channel between two general-purpose LLMs at matched scale.

Multimodal coupling. LLaVA [14], BLIP-2 [12], Flamingo [2], MiniCPM-o [16] couple vision and language by prepending projected visual tokens to an LLM’s input. BLIP-2 ablates query-token count, the closest prior analog to our N sweep. Our v2 transfers this pattern to language-language fast/slow coupling.

Continuous chain-of-thought. COCONUT [9], Pause Tokens [8], Implicit CoT [6], Quiet-STaR [21] all reason in latent space within a single model. Our work is the cross-model analog: two frozen models coupled by a learned latent channel.

Hierarchical / cascades. Speculative decoding [11] couples draft+verifier within one decoding loop; Mixture-of-Depths [19] adapts compute per token. These optimize a single forward pass; we decouple two models in wall-clock and pass the slower’s deliberation forward as a learned signal.

Position. To our knowledge this is the first reported head-to-head text-vs-latent comparison between two frozen general-purpose multimodal LLMs at matched scale, with deployment-time evaluation (not just offline KL) and a controlled negative domain (MetaDrive, where neither bridge helps) that bounds the claim.

10 Conclusion

A continuous latent bridge between two frozen multimodal LLMs at matched scale (9 B fast, 8 B slow) beats the text bridge by 26–82% on 4 of 7 evaluable Atari games under a fixed greedy decoder. We show this advantage is *decoder-specific*—it disappears at every fixed sampling temperature because the latent’s thin per-emission signal only sharpens the greedy argmax (§4.3, §4.4). The right comparison treats the decoder as a tuned deployment hyperparameter: with each channel at its own best decoder, selected on held-out seeds, the latent is **never significantly worse than the text bridge and is significantly better on 2 of 7 games** (MsPacman, RoadRunner; the other 5 are statistical ties), and the two channels demonstrably prefer different decoders (Table 2)—so the latent is a safe-or-better drop-in for the text channel, non-inferior everywhere and materially better where slow reasoning helps most, rather than a uniform winner at any single fixed decoder. Feeding *both* channels to the frozen head at once does not combine their strengths—it interferes, never beating the better single channel and significantly hurting 3 of 7 games (RoadRunner catastrophically, −96%; §4.5)—so the design rule is to couple via exactly one channel. Most “bridge collapsed” cells trace to Stage A action-head failures, chiefly OOD-brittleness under suffix/bridge inputs, and a targeted retraining fix recovers the largest single $L-T$ gap in the sweep (§5). The obvious objections do not hold: giving T more text widens the gap rather than closing it (§6.2), and replacing the trained latent with zero or random tokens costs most of its advantage where it wins (§6.3).

Rather than a “text is bandwidth-limited” story—which our own ablations do not support (§6.1, Appendix N)—we land on a sharper claim: **the latent bridge helps if and only if slow reasoning helps the task**. $L-F$ tracks $T-F$ at $r = 0.92$ across 7 Atari games and a driving domain ($r = 0.94$ over all 16 evaluated cells); MetaDrive is a controlled negative where the slow model never beats fast-only and the latent is correspondingly inert; and the per-game learned-content verdicts track the same variable (§7).

The cleanest follow-ups are (i) an emission-discrimination auxiliary loss during Stage C to push more per-tick state into the bridge—the trained latent is non-lexical and strongly game-conditioned but only weakly per-emission-conditioned, so its used capacity is far below its nominal 8×4096 (§6.4)—and (ii) testing the comparison in the production settings (voice agents, robotics) where the text channel actually lives.

References

- [1] Michael Ahn, Anthony Brohan, Noah Brown, et al. Do as i can, not as i say: Grounding language in robotic affordances. In *CoRL*, 2022.
- [2] Jean-Baptiste Alayrac et al. Flamingo: a visual language model for few-shot learning. In *NeurIPS*, 2022.
- [3] Ankesh Anand, Evan Racah, Sherjil Ozair, Yoshua Bengio, Marc-Alexandre Côté, and R Devon Hjelm. AtariARI: Atari annotated RAM interface, 2019. <https://github.com/mila-iqia/atari-representation-learning>.
- [4] Marc G Bellemare, Yavar Naddaf, Joel Veness, and Michael Bowling. The arcade learning environment: An evaluation platform for general agents. *Journal of Artificial Intelligence Research*, 47:253–279, 2013.

- [5] Anthony Brohan, Noah Brown, Justice Carbajal, et al. RT-2: Vision-language-action models transfer web knowledge to robotic control. In *CoRL*, 2023.
- [6] Yuntian Deng, Yejin Choi, and Stuart Shieber. From explicit CoT to implicit CoT: Learning to internalize CoT step by step, 2024. arXiv:2405.14838.
- [7] Google. Gemini Live multimodal real-time api, 2024. <https://ai.google.dev/gemini-api/docs/live>.
- [8] Sachin Goyal, Ziwei Ji, Ankit Singh Rawat, Aditya Krishna Menon, Sanjiv Kumar, and Vaishnavh Nagarajan. Think before you speak: Training language models with pause tokens. In *ICLR*, 2024.
- [9] Shibo Hao, Sainbayar Sukhbaatar, DiJia Su, Xian Li, Zhiting Hu, Jason Weston, and Yuan-dong Tian. Training large language models to reason in a continuous latent space, 2024. arXiv:2412.06769.
- [10] Wenlong Huang, Fei Xia, Ted Xiao, et al. Inner monologue: Embodied reasoning through planning with language models. In *CoRL*, 2022.
- [11] Yaniv Leviathan, Matan Kalman, and Yossi Matias. Fast inference from transformers via speculative decoding. In *ICML*, 2023.
- [12] Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *ICML*, 2023.
- [13] Quanyi Li, Zhenghao Peng, Lan Feng, Qihang Zhang, Zhenghai Xue, and Bolei Zhou. MetaDrive: Composing diverse driving scenarios for generalizable reinforcement learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(3):3461–3475, 2023.
- [14] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. In *NeurIPS*, 2023.
- [15] OpenAI. OpenAI Realtime API, 2024. <https://platform.openai.com/docs/guides/realtime>.
- [16] OpenBMB Team. MiniCPM-o 4.5: An omni-modal large language model, 2025. https://huggingface.co/openbmb/MiniCPM-o-4_5.
- [17] Qwen Team. Qwen3-VL-8B-Thinking, 2025. <https://huggingface.co/Qwen/Qwen3-VL-8B-Thinking>.
- [18] Antonin Raffin. Stable-Baselines3 atari zoo, 2021. <https://huggingface.co/sb3>.
- [19] David Raposo, Sam Ritter, Blake Richards, et al. Mixture-of-depths: Dynamically allocating compute in transformer-based language models, 2024. arXiv:2404.02258.
- [20] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms, 2017. arXiv:1707.06347.
- [21] Eric Zelikman, Georges Harik, Yijia Shao, Varuna Jayasiri, Nick Haber, and Noah D Goodman. Quiet-STaR: Language models can teach themselves to think before speaking. In *COLM*, 2024.

A Full results table

Both bare-Stage-A and robust-Stage-A reported for every game, $n = 12$ per cell.

Game (variant)	F	T	L	Δ_{L-T} (% , p)
MsPacman	256 ± 25	408 ± 92	628 ± 356	+54 % ($p=0.06$)
MsPacman (robust SA)	325 ± 72	61 ± 3	60 ± 0	-1 % ($p=0.36^\dagger$)
Seaquest	42 ± 20	63 ± 12	80 ± 0	+26 % ($p<.001^\dagger$)
Seaquest (robust SA)	20 ± 0	0 ± 0	0 ± 0	—
RoadRunner	0 ± 0	608 ± 250	967 ± 49	+59 % ($p<.001$)
RoadRunner (robust SA)	958 ± 67	1000 ± 0	925 ± 45	-8 % ($p<.001^\dagger$)
River Raid	1067 ± 88	383 ± 60	360 ± 0	-6 % ($p=0.01^\dagger$)
River Raid (robust SA)	1032 ± 20	337 ± 81	612 ± 310	+82 % ($p=0.01$)
Enduro	3 ± 3	0 ± 0	8 ± 9	—
Enduro (robust SA)	1 ± 1	5 ± 6	6 ± 3	+19 % ($p=0.63$)
Q*bert	25 ± 0	0 ± 0	0 ± 0	—
Q*bert (robust SA)	25 ± 0	125 ± 0	50 ± 0	-60 % ($p<.001^\dagger$)
SpaceInvaders	105 ± 0	0 ± 0	0 ± 0	—
SpaceInvaders (robust SA)	107 ± 62	18 ± 19	15 ± 0	-18 % ($p=0.27^\dagger$)
Pong	-21 ± 0	-21 ± 0	-21 ± 0	+0 % ($p=1.00^\dagger$)

Reading the table. Welch’s t becomes undefined or unreliable when either cell has zero variance; in those cases we report Mann–Whitney U (marked †). On cells where *both* sides have zero variance (e.g., Q*bert/robust $T = 125 \pm 0$, $L = 50 \pm 0$), the test reduces to comparing two integers; the p -value is mechanical, not a sampling statement.

B Statistics methodology

Variance sources. Each cell is $3 \text{ seeds} \times 4 \text{ episodes per seed} = 12 \text{ episodes}$. Seeds vary the ALE initial state. Action selection is greedy (argmax over logits). For games with small action spaces and policies that consistently pick the same action sequence, every episode within a seed often hits the same trajectory; hence many cells have $\sigma = 0$. This is not a sampling artifact—it is the policy being deterministic given the env seed.

When we trust which test.

- Both cells have $\sigma > 0$ and roughly normal: **Welch’s t** .
- One cell has $\sigma = 0$, the other has support: **Mann–Whitney U** .
- Both cells have $\sigma = 0$: **we report the comparison but do not claim a p -value** (the test is deterministic-trajectory difference, not a population inference).

Test convention. All Welch’s t values use the unbiased sample-variance form, as in SciPy’s `ttest_ind` with `equal_var=False`.

Bootstrap CIs. 10^4 resamples for each cell mean.

Cohen’s d . Reported only when both cells have $\sigma > 0$. Zero-variance cells produce mechanically large d that we do not interpret.

C Slow emission samples

RoadRunner.

“Got it, let’s break this down. The current state: Road Runner is at (3,0), Coyote is at (0,0) with a Δ_x of +3, so Coyote is closing in. The nearest birdseed is at $x = 159$, which is ahead. There are obstacles at (170,16) and (170,0), so trucks/landmines are in the way. . . .”

Q*b_{ert}.

*“Got it, let’s analyze the Q*b_{ert} state. The player is at (0,0), Coily is at (65,0) which is on the right edge. The purple and green balls are at $y = 148$, so they’re falling. The goal is to jump tiles to change colors, avoid Coily and balls. First, threat: Coily is on the right edge, so if Q*b_{ert} moves. . . .”*

Both samples are ~ 300 characters. RoadRunner’s encodes simultaneous Δ_x for Coyote, target x for birdseed, two obstacle coordinates, and a directional commitment. Q*b_{ert}’s encodes (jump-direction, target-color, threat-actor)—a triple. This is the qualitative content that motivated the lexical-diversity hypothesis tested (and rejected) in §7.4; we make no quantitative prediction on top.

D Stage A and Stage C details

Stage A. AdamW, lr 1×10^{-4} , 3 epochs, batch=1 with grad-accum=8. The action head has 18 outputs (full ALE action space) on every game; the random baseline below is $1/k$ for k = the game’s ALE *minimal* action set (the only actions the SB3 expert emits, so the val-set classes). Bare val-accuracies on the game-state-only prompt: MsPacman 32% (random 11.1%, $k=9$), Seaquest 24% (5.6%, $k=18$), SpaceInvaders 33% (16.7%, $k=6$), Pong 25% (16.7%, $k=6$), RoadRunner 59% (5.6%, $k=18$), River Raid 31% (5.6%, $k=18$), Enduro 49% (11.1%, $k=9$), Q*b_{ert} 34% (16.7%, $k=6$). Robust Stage A is identical except half of training batches receive a synthetic slow-style text suffix appended (suffix-prob=0.5); bare val-acc barely changes (e.g., MsPacman 32 $\rightarrow$ 33%, SI 33 $\rightarrow$ 30%) but the policy is suffix-invariant.

Stage C. 2-layer MLP projection (LayerNorm, Linear 4096 $\rightarrow$ 4096, GELU, Linear 4096 $\rightarrow$ 4096, LayerNorm), 33.6 M parameters. AdamW, lr 5×10^{-5} , 1 epoch, batch=1 with grad-accum=4. KL temperature $\tau = 1.0$. Slow residuals extracted at layer 24 of 36 (we did not ablate the layer choice). Default $N = 8$ latent tokens, $\sim 5K$ (frame, slow text, slow residuals) tuples per game from Stage B random-policy rollouts.

Text-bridge suffix. The slow’s full unmodified post-thinking emission, appended verbatim under a “strategic-guidance” tag; no truncation; median 302 chars across games.

Slow prompt template (excerpt, RoadRunner, paraphrased). The object coordinates fed to the slow model are extracted from the ALE RAM in the style of the Atari Annotated RAM Interface [3]. “Road Runner is at (x_{rr}, y_{rr}) , Coyote at (x_c, y_c) with $\Delta_x = +3$. Nearest birdseed at $x = 159$; obstacles at (170,16) and (170,0). . . .” The system prompt instructs the slow to emit 1–3 sentences identifying threat/opportunity and recommended intent—*not* individual actions.

E Latency breakdown

Per-tick wall-clock on a contended-GPU host (vLLM workers sharing the same H100 family):

	F (no cache)	F (vrf=4)	L (vrf=4)
Vision tower	70 ms	17.5 ms (amortized)	17.5 ms
LLM prefill	60 ms	13 ms	18 ms (+8 latent tokens)
LLM decode (1 action token)	27 ms	2.5 ms	2.5 ms
Per-tick total	~ 157 ms	~ 33 ms	~ 38 ms

The bridge itself adds only ~ 5 ms over F . End-to-end “ $L \approx 124$ ms” figures from earlier work also include async-callback completion time under GPU contention. Vision-token caching across N ticks

(vrf) reduces vision-tower cost by $N\times$ at the cost of perception staleness; score is non-monotonic in vrf (vrf=4: $F=110$, vrf=15: $F=140$ on MsPacman), an interaction of staleness with multi-tick action commitments. (These are a separate latency-stress configuration with caching enabled and so do not match the headline MsPacman $F=256$, which uses per-tick vision.)

F Bridge-token diagnostics (full table)

Backing data for §6.4. Bridge tokens computed by loading the trained Stage C v2 projection for each game and applying it to that game’s cached seed-0 slow residuals; up to 8 emissions per game.

Norms. Vocab embeddings: mean 1.453, std 0.359, p1=0.281, p99=1.923. Bridge tokens: norm ≈ 64 across all 7 games (std ≤ 0.01 within a game)—fixed by the output LayerNorm of the projection MLP.

Nearest-vocab cosine, averaged over the 8 bridge positions of one emission:

Game	avg max cosine to vocab	avg p99.9 cosine to vocab
MsPacman	0.076	0.053
RoadRunner	0.081	0.058
Q*bert	0.087	0.058
Seaquest	0.067	0.044
River Raid	0.074	0.050
SpaceInvaders	0.070	0.047
Enduro	0.063	0.034

For reference, two random vocab embeddings have cosine ≈ 0.04 . Every bridge token is essentially at that random-vocab background. The “top-k tokens” from a plug-in nearest-neighbor decode are therefore the maximum of a uniform-low distribution, not real semantic matches.

Cross-game cosine (mean pairwise bridge similarity over $E \times N$ tokens):

	MsPac	RR	Q*b	Seaq	RivR	SI	Endu
MsPacman	+0.87	+0.18	+0.13	+0.21	+0.10	+0.25	+0.04
RoadRunner	+0.18	+0.81	+0.13	+0.21	+0.09	+0.15	+0.10
Q*bert	+0.13	+0.13	+0.81	+0.18	+0.15	+0.14	+0.04
Seaquest	+0.21	+0.21	+0.18	+0.80	+0.11	+0.23	+0.09
River Raid	+0.10	+0.09	+0.15	+0.11	+0.89	+0.14	+0.06
SpaceInvaders	+0.25	+0.15	+0.14	+0.23	+0.14	+0.85	+0.06
Enduro	+0.04	+0.10	+0.04	+0.09	+0.06	+0.06	+0.86

Diagonals 0.80–0.89; off-diagonals 0.04–0.25. The bridge is strongly game-conditioned.

Within-game across-emission consistency (cosine between per-emission mean bridge tokens, off-diagonal):

Game	off-diag mean	off-diag std
MsPacman	+0.951	0.019
RoadRunner	+0.938	0.027
Q*bert	+0.918	0.025
Seaquest	+0.920	0.032
River Raid	+0.968	0.019
SpaceInvaders	+0.961	0.032
Enduro	+0.956	0.027

Within-game per-emission variation is small ($1-\bar{c} \approx 3-8\%$), suggesting most of the latent capacity is used to encode the (fixed-within-episode) game identity, with a thin per-tick residual encoding strategic state. This is the basis for the “used capacity $\ll$ nominal” observation in §6.4.

Legacy MI probe (kept for completeness). A pre-revision binned plug-in MI estimate on the trained MsPacman bridge gave $I(\text{bridge}; \text{sign of future-30-tick reward}) = 0.024$ nats vs a shuffled baseline of 0.014 nats ($+0.010, 1.4\sigma$). Action MI was at baseline (0.087 vs 0.085, training used random-policy trajectories). We do not draw conclusions from these numbers; the diagnostics above are more informative.

G OOD KL probe: full numbers

Per-emission probe: for each (game, Stage A variant), we walk the cached seed-0 T -trajectory, take up to 20 emissions, and at each emission tick compute $\text{KL}(p_{\text{bare-prompt}} \parallel p_{\text{suffixed-prompt}})$ over legal actions, the argmax-change rate $\mathbf{1}[\text{argmax } p_{\text{bare}} \neq \text{argmax } p_{\text{suff}}]$, and the probability mass on the bare argmax under each prompt. The *collapse?* column marks the three games whose bare-Stage-A bridges were degraded or zeroed at deployment (SpaceInvaders, Q*bert, River Raid; the “collapsed” group of §5).

Bare Stage A:

Game	n	KL	argmax-change	$p_{\text{bare}}(a_{\text{bare}}^*)$	$p_{\text{suff}}(a_{\text{bare}}^*)$	collapse?
MsPacman	20	0.54 ± 0.47	0.85	0.55	0.19	no
Seaquest	20	1.00 ± 0.28	0.70	0.29	0.10	no
RoadRunner	8	0.39 ± 0.14	0.75	0.52	0.24	no
SpaceInvaders	19	0.29 ± 0.15	0.79	0.49	0.24	yes
Q*bert	20	0.25 ± 0.22	0.90	0.41	0.20	yes
River Raid	20	0.24 ± 0.06	0.70	0.24	0.11	yes
<i>group means: non-collapsed</i>		0.64	0.77	0.45	0.17	
<i>group means: collapsed</i>		0.26	0.80	0.38	0.18	

Robust Stage A (suffix-prob=0.5):

Game	n	KL	argmax-change	$p_{\text{bare}}(a_{\text{bare}}^*)$	$p_{\text{suff}}(a_{\text{bare}}^*)$	collapse?
MsPacman	20	0.40 ± 0.23	0.30	0.42	0.52	no
Seaquest	20	0.34 ± 0.19	0.90	0.30	0.13	no
RoadRunner	8	0.11 ± 0.07	0.12	0.68	0.59	no
SpaceInvaders	19	0.19 ± 0.08	0.89	0.37	0.19	yes
Q*bert	20	0.11 ± 0.07	0.50	0.36	0.33	yes
River Raid	20	0.26 ± 0.13	0.65	0.24	0.20	yes
<i>group means: non-collapsed</i>		0.28	0.44	0.47	0.41	
<i>group means: collapsed</i>		0.19	0.68	0.32	0.24	

Read. (1) Under bare Stage A, the suffix flips the argmax on 70–90 % of emissions on *every* game; the OOD effect is large and uniform. (2) KL is *lower* on the collapsed games (0.26 vs 0.64), against the naive prediction. This is because the bare head on collapsed games is less peaky to start with, so the same argmax flip moves less probability mass. KL is therefore not the right summary statistic for the OOD effect—argmax-change is. (3) Robust Stage A cuts argmax-change cleanly on two of the three games where bare was already non-degenerate (RoadRunner 75→12, MsPacman 85→30; Seaquest is the exception at 70→90, matching its deployment harm under robust Stage A), confirming the fix targets the right thing where it works. On collapsed games robust Stage A reduces KL but only partially reduces argmax-change (Q*bert 90→50; River Raid 70→65), consistent with the asymmetric rescue we observed in scoring (§5).

H Decoder ablation on Q*bert (full numbers)

Backing data for §4.3. Robust Stage A, $n=12$ episodes per cell (3 seeds $\times$ 4 episodes), ≤ 500 ticks per episode, multinomial sampling from $\text{softmax}(\text{logits}/\tau)$ over legal actions.

Decoder	Strategy	mean $\pm$ std	median	min	max
greedy (paper)	F	25.0 $\pm$ 0.0	25.0	25	25
greedy (paper)	T	125.0 $\pm$ 0.0	125.0	125	125
greedy (paper)	L	50.0 $\pm$ 0.0	50.0	50	50
sample, $\tau = 0.5$	F	16.7 $\pm$ 11.8	25.0	0	25
sample, $\tau = 0.5$	T	139.6 $\pm$ 66.5	125.0	50	250
sample, $\tau = 0.5$	L	125.0 $\pm$ 66.9	125.0	50	250
sample, $\tau = 1.0$	F	66.7 $\pm$ 71.7	50.0	0	225
sample, $\tau = 1.0$	T	87.5 $\pm$ 65.0	75.0	25	250
sample, $\tau = 1.0$	L	250.0 $\pm$ 155.8	200.0	50	650

Significance tests, L vs T :

- $\tau=0.5$: Welch $t = 0.51$, $p = 0.61$ (n.s.). $L \approx T$.
- $\tau=1.0$: Welch $t \approx 3.2$, $p \approx 0.006$. L beats T by $2.9\times$.

Reading. Under greedy, every episode is deterministic given the env seed; with a small action space and consistent argmax, all 12 episodes per cell hit the same trajectory (std=0). The bridge’s small per-emission variance (cosine 0.92–0.97 between emissions, Appendix F) is below the threshold needed to flip an argmax. Multinomial sampling exposes this variance: as τ increases, L ’s mean grows (50 $\rightarrow$ 125 $\rightarrow$ 250) while T ’s plateaus or declines (125 $\rightarrow$ 140 $\rightarrow$ 88). The bridge’s per-tick information advantage over text is therefore real but *not visible* under greedy decoding on Q*bert.

I Longer T-suffix ablation (full numbers)

Backing data for §6.2. Same models, Stage A bare checkpoints, $n=12$ per cell (3 seeds $\times$ 4 episodes), ≤ 500 ticks per episode. “Window” is the number of most-recent slow emissions concatenated into the T suffix (separated by a single space); the bridge L is unchanged.

Game	Window	mean $\pm$ std	median	min	max
MsPacman	$w = 1$ (paper)	407.5 $\pm$ 92.2	385.0	270	600
MsPacman	$w = 2$	378.3 $\pm$ 117.5	375.0	130	680
MsPacman	$w = 3$	351.7 $\pm$ 87.7	335.0	240	560
RoadRunner	$w = 1$ (paper)	608.3 $\pm$ 250.3	650.0	0	900
RoadRunner	$w = 2$	8.3 $\pm$ 27.6	0.0	0	100
RoadRunner	$w = 3$	0.0 $\pm$ 0.0	0.0	0	0

Reading. On MsPacman, doubling/tripling the T budget gives a small monotonic decline (–14% from $w=1$ to $w=3$). On RoadRunner, $w=2$ collapses to near-zero and $w=3$ collapses to exact zero on every episode. Older emissions appear to be stale and to mislead the action head; the fast model does not learn an implicit time-ordering of the concatenated emissions. The L scores stay at their paper values ($L_{\text{MsPacman}}=628$, $L_{\text{RoadRunner}}=967$), so the $L-T$ gap widens monotonically with w .

J Sampling on other greedy-degenerate games (full numbers)

Backing data for the “greedy-decoding bias is general” claim in §4.3. Action-policy=sample at $\tau = 1.0$, $n = 12$ per cell, ≤ 500 ticks per episode. The variant is robust Stage A except for the **Seaquest (bare)** block, which is the *headline* Seaquest variant (§4) and the most consequential row: its greedy $L > T$ win does not survive sampling.

Game (variant)	Strategy	Greedy (paper)	Sample $\tau = 1.0$
Seaquest (bare)	F	41.7 ± 19.9	63.3 ± 35.0
Seaquest (bare)	T	63.3 ± 11.5	143.3 ± 37.0
Seaquest (bare)	L	80.0 ± 0.0	135.0 ± 40.1
SpaceInvaders (robust)	F	107.1 ± 62.4	106.7 ± 26.0
SpaceInvaders (robust)	T	18.3 ± 18.6	107.5 ± 19.2
SpaceInvaders (robust)	L	15.0 ± 0.0	116.7 ± 14.5
Seaquest (robust)	F	20.0 ± 0.0	110.0 ± 38.7
Seaquest (robust)	T	0.0 ± 0.0	41.7 ± 37.8
Seaquest (robust)	L	0.0 ± 0.0	33.3 ± 27.5
Enduro (robust)	F	0.8 ± 1.1	3.2 ± 2.5
Enduro (robust)	T	4.9 ± 5.9	0.2 ± 0.8
Enduro (robust)	L	5.8 ± 2.6	0.6 ± 1.4

Reading. Seaquest (bare) is the consequential row: under greedy $L = 80 \pm 0 > T = 63$ (a +26% headline win), but the L cell is deterministic; under sampling both bridges roughly double off F ($F: 42 \rightarrow 63$, $T: 63 \rightarrow 143$, $L: 80 \rightarrow 135$) and the ordering flips to $L \approx T$ with text nominally ahead (Welch $p = 0.60$, MWU $p = 0.73$). The slow channel still clearly helps ($T, L \gg F$, $p < 10^{-3}$), but the latent-over-text margin was a greedy artifact—greedy had under-counted T here more than L , the opposite of Q*bert. SpaceInvaders is the clearest example of greedy under-counting the bridge: under greedy both bridges are at ~ 15 – 18 (far below $F=107$), suggesting bridge-induced collapse; under sampling all three strategies cluster around 107–117. The bridges were never broken on SpaceInvaders—greedy was misrepresenting them. Seaquest/robust shows the same pattern less dramatically: $T=L=0$ (greedy) $\rightarrow 33$ – 42 (sampling); the bridges are still below $F=110$ but no longer collapsed. Enduro inverts the pattern: under sampling the bridges *drop* ($T 5 \rightarrow 0.2$, $L 6 \rightarrow 0.6$), so Enduro’s greedy advantage was real but fragile. The general lesson is that greedy decoding at small action-space scale produces deterministic trajectories that can mask both the bridge’s signal and the bridge’s actual failure modes *and* the sign of the L -vs- T comparison; comparing F/T/L should be done under a sampling decoder when per-episode variance is required for meaningful statistics.

K Full decoder sweep (greedy / $\tau \in \{0.3, 0.5, 0.7, 1.0, 1.5\}$)

Backing data for §4.4. We re-ran the reported-variant cells at greedy and five sampling temperatures on the *same* Stage A/C checkpoints, $n = 12$ (3×4), ≤ 500 ticks. Greedy reproduces the headline exactly on MsPacman, River Raid, and Seaquest, and F reproduces on every game (it never invokes the slow model)—so the greedy-vs-sampling difference is a genuine decoder effect, not checkpoint or code drift. The per-decoder means:

Game	decoder	F	T	L	L vs T
MsPacman	greedy	256	408	628	$L > T$
	$\tau=0.3$	268	343	353	$L > T$
	$\tau=0.5$	358	440	360	$L < T$
	$\tau=0.7$	347	342	339	$L < T$
	$\tau=1.0$	348	292	368	$L > T$
RoadRunner	greedy	0	475	608	$L > T$
	$\tau=0.3$	0	142	17	$L < T$
	$\tau=0.5$	0	8	0	$L < T$
	$\tau=0.7$	0	17	0	$L < T$
	$\tau=1.0$	0	25	17	$L < T$
River Raid	greedy	1032	337	607	$L > T$
	$\tau=0.3$	946	668	511	$L < T$
	$\tau=0.5$	1015	717	562	$L < T$
	$\tau=0.7$	890	646	641	$L < T$
	$\tau=1.0$	867	649	531	$L < T$
Seaquest	greedy	42	63	80	$L > T$
	$\tau=0.3$	50	125	130	$L > T$
	$\tau=0.5$	48	128	122	$L < T$
	$\tau=0.7$	58	127	137	$L > T$
	$\tau=1.0$	nan	nan	nan	$L = T$
	$\tau=1.5$	63	113	97	$L < T$
Q*bert	greedy	25	125	50	$L < T$
	$\tau=0.3$	19	110	148	$L > T$
	$\tau=0.5$	19	200	162	$L < T$
	$\tau=0.7$	38	138	123	$L < T$
	$\tau=1.0$	108	177	188	$L > T$
	$\tau=1.5$	88	242	169	$L < T$
Enduro	greedy	3	3	2	$L < T$
	$\tau=0.5$	4	3	1	$L < T$
	$\tau=1.0$	3	0	1	$L > T$
SpaceInvaders	greedy	107	18	15	$L < T$
	$\tau=0.5$	115	104	109	$L > T$
	$\tau=1.0$	135	162	142	$L < T$

The $L > T$ advantage is greedy-specific. On every game, the greedy $L > T$ margin vanishes under *any* fixed sampling temperature: $L \approx T$ (MsPacman, Seaquest) or significant $T > L$ (River Raid at $\tau=1.0$; RoadRunner collapses to the $F=0$ floor). The latent’s small per-emission variance (§6.4) shifts the deterministic argmax but is swamped once the decoder samples, so the latent yields a better *greedy policy* rather than a more robustly informative channel.

RoadRunner is also hardware-fragile. RoadRunner greedy does *not* reproduce the paper’s headline ($L=967$, run on an H100 family per Appendix E): on the Blackwell GPU used for these re-runs it gives a tight $L=608$. Because F reproduces exactly and only the slow-model-dependent cells drift, and only on the one game whose $F=0$ makes the score hinge entirely on the slow emission triggering one precise action, the cause is floating-point nondeterminism in the (greedy, `do_sample=False`) slow-model generation across hardware. The $L > T$ *direction* is robust (608 vs 475); the headline magnitude is hardware-specific.

L Held-out decoder selection protocol

For the best-achievable comparison (§4.4, Table 2) we treat the action decoder as a deployment hyperparameter and select it per (game, channel) the honest way—tune on held-in data, report on

held-out—using leave-one-seed-out over the 3 ALE seeds (4 episodes each):

- For each held-out seed s : pick the decoder with the highest mean over the *other* two seeds (8 episodes), then record that decoder’s 4 episodes on seed s .
- The reported best-achievable cell pools the 3 held-out folds (12 episodes, each seed held out once).
 L vs T uses Welch’s t and Mann–Whitney on the two pooled held-out vectors.

This is an unbiased estimate of “tune the decoder, then deploy,” unlike an oracle max over decoders on the same episodes (which we report alongside as an optimistic upper bound). In practice the selection is stable—held-out $\approx$ oracle on most cells (e.g. MsPacman- L picks greedy on all three folds)—so the unbiased estimate sits close to the optimistic one. The procedure is in `scripts/decoder_select_cv.py`; the table is generated by `scripts/make_best_achievable_table.py`.

M Bridge replacement control (full numbers)

Backing data for §6.3. MsPacman bare, Stage A bare checkpoint, $n = 12$ per cell, greedy decoding (paper default). Three L variants compared against F : “zero” replaces the 8 trained bridge tokens with 8 zero vectors, “random” replaces them with random Gaussian vectors rescaled to per-position L2 norm ≈ 64 (matching trained), “trained” is the unmodified projection output.

Variant	mean $\pm$ std ($n = 12$)	median	vs F	p (Welch, vs trained)
F (no bridge)	256 ± 25	250	—	—
L_{zero}	379 ± 95	395	+48%	0.04
L_{random}	387 ± 161	365	+51%	0.05
L_{trained}	628 ± 356	550	+145%	—

Reading. L_{zero} vs L_{random} : Welch $t=0.15$, $p=0.88$ (n.s.)—essentially the same. So the architectural lift over F comes from *having 8 prepended slots*, not from the slots having any particular content. L_{trained} vs L_{random} : Welch $t \approx 2.12$, $p \approx 0.05$ (vs L_{zero} : $t \approx 2.34$, $p \approx 0.04$). The trained content is contributing a real signal, on top of the architectural lift. Decomposition on MsPacman: $\sim 40\%$ of L ’s advantage over F is architectural; $\sim 60\%$ is learned bridge content.

All-games sweep (the control run on every Atari game, not just MsPacman). The verdict table below is summarized in the main text (§7.3). We re-ran the trained/zero/random control on all 7 games using the *bare* Stage A head under greedy decoding—the natural common setting for a content control—with the single exception of Q*bert, run on its robust head under $\tau=1$ sampling (its canonical decoder, since greedy Q*bert is degenerate, §4.3); $n = 12$ per cell (3×4). L_{random} uses matched per-position L2 norm. Because the control is run *bare*, the $T-F$ column below is the bare-head value, so three games (Enduro, SpaceInvaders, River Raid) use their bare cell here even though the headline reports their robust variant—the learned-vs-control verdict does not depend on that choice. This is also an independent re-run, so its MsPacman cell ($L_{\text{trained}}=666$) and Seaquest cell (100) differ slightly from the standalone / headline runs above (628 and 80); all agree within run-to-run noise. The Q*bert cell ($L_{\text{trained}}=123$) likewise differs from the decoder ablation’s $\tau=1$ mean (250 ± 156 , Appendix H); both are single high-variance sampling runs and the gap is within one standard deviation, but the verdict here (trained $\approx$ random) does not depend on the absolute value. “Learned” = L_{trained} exceeds $\max(L_{\text{zero}}, L_{\text{random}})$ by $> 10\%$.

Game	L_{train}	L_{zero}	L_{rand}	verdict	$T-F$
RoadRunner	967	0	8	learned (Welch $t=58$)	+608
Seaquest	100	28	5	learned ($t=36$)	+22
MsPacman	666	408	410	learned ($t=2.3$)	+152
Enduro	7.8	1.4	4.7	trend only (tiny, $t=1.1$)	−3
Q*bert	123	63	117	$\approx$ random (tie)	+100
SpaceInvaders	0	148	90	<i>harmful</i>	−105
River Raid	360	1013	1003	<i>harmful</i>	−683

Reading: learned content tracks the predictor. The trained latent carries genuinely learned, behavior-relevant content— $L_{\text{trained}} \gg \max(L_{\text{zero}}, L_{\text{random}})$ —on **RoadRunner, Seaquest, and MsPacman**, exactly the games where slow reasoning helps ($T > F$); on RoadRunner the latent is *almost entirely* learned (controls drop to the floor, 0 and 8). On the games where slow reasoning does not help ($T \leq F$: River Raid, SpaceInvaders), the trained latent is *harmful*—zeroing or randomizing it scores *better*—the same inert/harmful pattern as MetaDrive (§7.1). Two off-diagonal cases are benign: Enduro’s scores are too small for the comparison to be meaningful ($t=1.1$, n.s.), and Q*bert—the one decoder-fragile game (§4.3)—shows trained $\approx$ random under sampling, consistent with its thin per-emission signal. So the MsPacman “ $\sim 60\%$ learned / $\sim 40\%$ architectural” decomposition is *not* universal: the learned fraction ranges from $\sim 100\%$ (RoadRunner) to negative (River Raid), and its sign is predicted by whether $T > F$. This is the mechanistic complement to the $r=0.92$ predictor: the latent helps where—and only where—it has carried real content from a slow channel that itself beats fast-only.

N 30B-A3B cross-scale ablation (full numbers)

Replicates the MsPacman/bare F/T/L cells with the slow model swapped from Qwen3-VL-8B-Thinking (dense) to Qwen3-VL-30B-A3B-Thinking (MoE, $\sim 3\text{B}$ active per forward, 48 layers, hidden_dim=2048). Pipeline:

- **Stage B:** 15 T -trajectories collected with the 30B slow (~ 470 emissions, ~ 7200 ticks). Same prompt template as 8B.
- **Stage C v2:** re-trained projection $2048 \rightarrow 4096$ ($\sim 33\text{M}$ parameters, same 2-layer MLP topology), 1 epoch over $\sim 7\text{K}$ samples. Final mean KL = 0.015 (*vs* 8B’s ~ 0.005 ; slightly under-converged given $\sim 3\times$ less data per slow-model parameter).
- **Eval:** F/T/L on MsPacman, 6 seeds $\times$ 2 episodes per cell, $n = 12$, ≤ 500 ticks/episode. Run twice: greedy and sample $\tau=1.0$.

Configuration	F	T	L	Δ_{L-T}
8B-dense, greedy (paper)	256 ± 25	408 ± 92	628 ± 356	+54% (Welch $p=0.06$)
30B-A3B, greedy, $n = 12$	273 ± 45	312 ± 14	210 ± 0	−33% (det. artifact, Q*bert-style)
30B-A3B, sample $\tau = 1.0$, $n = 12$	309 ± 81	389 ± 209	458 ± 283	+18% (Welch $p=0.50$, $d = 0.28$)

Reading. A “bigger slow model carries more, so $L-T$ should grow with capacity” hypothesis is not supported here. With Qwen3-VL-30B-A3B as slow, the $L > T$ direction is preserved under sampling (+18%) but the magnitude *shrinks* relative to 8B-dense (+54%), and significance is lost ($p=0.50, n=12$). The result is consistent with several non-exclusive readings:

- **Active parameters, not total.** 30B-A3B activates only $\sim 3\text{B}$ of its 30B parameters per forward pass. A dense-30B comparison (which we cannot run on this GPU) would be the right capacity test.
- **Hidden-dim shift.** 30B-A3B has hidden_dim=2048 vs 8B’s 4096, so the trainable projection maps $2048 \rightarrow 4096$ rather than $4096 \rightarrow 4096$. The same parameter budget covers a harder map.
- **Bridge under-training.** We trained on ~ 470 30B emissions; the 8B paper baseline used $\sim 5\text{K}$. Final KL 0.015 vs 0.005 is consistent with partial convergence.
- **One-game evidence.** A single game on a single cross-scale data point is not strong enough to falsify the scaling claim; we report this as a single counter-data point, not as a refutation.

The greedy result ($L = 210 \pm 0$) reproduces the deterministic-trajectory pattern we documented for Q*bert under greedy in the main text—another case where greedy decoding interacts pathologically with the bridge’s low per-emission variance.

O Per-game collapse triage

Across the 14 (game, variant) cells in Appendix A, five have a strategy cell at zero (the first five rows below). Robust Stage A additionally destroys one already-working cell without driving it

to exactly zero (MsPacman/robust, $L \rightarrow 60$), and a separate game, Pong (outside the 14), sits at the random-policy floor (-21). Stage A val-accuracy is the load-bearing predictor of collapse, not bridge-training failure: σ of T, L at deployment correlates with bare Stage A val-acc, not with Stage C KL.

Game/variant	Collapsed cell	Diagnosis
RoadRunner/bare	$F = 0$	Stage A predicts NOOP; bridges rescue.
SpaceInvaders/bare	$T = L = 0$	Stage A OOD-brittle; robust partially rescues.
Q*bert/bare	$T = L = 0$	Stage A OOD-brittle; robust rescues to $T=125, L=50$.
Seaquest/robust	$T = L = 0$	Robust drops below bare F ; use bare.
Enduro/bare	$T = 0$	Text collapses; robust gives a non-degenerate near-floor tie.
MsPacman/robust	$L \rightarrow 60$	(non-zero) Robust destroys MsPacman policy; use bare.
Pong/all	all -21	(outside 14) Stage A val-acc only $1.5\times$ random; cannot be rescued.

P MetaDrive (non-Atari) full numbers

Setup. MetaDrive [13] at ~ 10 Hz control. The fast model sees a top-down rendered 84×84 image (rendered via the pygame top-down channel, no GL—this is what makes headless end-to-end runs feasible on one GPU); 9 discrete actions (3 steer $\times$ 3 throttle). The slow model sees the structured state (speed, heading, lane offset, upcoming-turn cue derived from `navi_arrow_dir`, nearby vehicles). Stage A clones MetaDrive’s built-in PPO [20] expert; Stage B collects slow emissions under expert driving; Stage C distills the latent against the (robust, suffix-aware) text teacher. Random policy scores ~ 8 reward; the discrete expert scores ~ 174 – 211 .

Two task regimes. *Reactive* = default 3-block map (lane-keeping dominates). *Planning* = map **SXSXSX** (straights + X-intersections) with off-route termination, so survival requires correct turn decisions: the route expert scores 211 ± 65 vs 93 ± 23 for a constant straight+throttle policy ($n=40$ episodes each), a $+118$ -reward gap. Eval on the robust head: $n=8$ (4 seeds $\times$ 2 episodes) for the planning rows, $n=6$ (3 seeds $\times$ 2) for the reactive row.

Regime	Decoder	F	T	L
Reactive	greedy	71.2	69.5	69.5
Planning	greedy	87.8	85.1	85.1
Planning	sample $\tau=1$	123.2	43.5	36.7

Bridge-replacement control (planning, greedy, $n=8$). $L = 85.1$, $L_{\text{zero}} = 89.5$, $L_{\text{random}} = 97.1$: replacing the trained latent with zeros or matched-norm random vectors does *not* reduce the score (if anything it rises), i.e. the latent is inert—it carries no behavior-relevant information. This is the opposite of MsPacman ($L_{\text{trained}} = 628 \gg L_{\text{random}} = 387$, §M), and is the diagnostic basis for calling MetaDrive a controlled negative.

Why L cannot exceed F here. Stage C trains L to match the text teacher’s action distribution (objective $\text{KL}(\pi_L \parallel \pi_T)$), so L ’s ceiling is T . On MetaDrive $T \leq F$ in every configuration we ran—both task regimes (reactive lane-keeping, planning intersections), both decoders (greedy, sampling), and both Stage A heads: the bare head, where the text channel collapses ($T=17.2$ vs $F=73.7$), and the robust head, where T sits just below F in the tabulated reactive (69.5 vs 71.2), planning-greedy (85.1 vs 87.8), and planning-sampling (43.5 vs 123.2) cells—so the best attainable latent is $L \approx F$. Fixing the teacher (robust, suffix-aware head, expert-driven data, KL converged to 0.004) makes the distillation *clean*— $L = T$ to the decimal under greedy—but cannot raise a ceiling set by a teacher that does not beat fast-only. The slow model’s frame-grounded, ~ 1.5 s reasoning simply does not produce better driving decisions than the reactive policy, even when the task demands route planning.

Robustness of the predictor. The $(T-F, L-F)$ correlation (Figure 6) is robust to both variant selection and individual games. *Variant selection:* $r = 0.92$ over the 8 reported-variant cells and $r = 0.94$ over all 16 (game, variant) cells with no selection—the 16 are the 14 bare/robust Atari cells, a SpaceInvaders expert-data cell (§5), and MetaDrive—so the choice of which Stage A variant to headline does not create the relationship. *Individual games* (on the 8 reported-variant cells): leave-one-out Pearson r stays in $[0.89, 0.98]$ for all 8 deletions (dropping the RoadRunner high-leverage point gives $r = 0.89$); the rank correlation is Spearman $\rho = 0.98$; a game-level bootstrap (10,000 resamples) gives a 95 % CI of $[0.81, 1.0]$; and excluding both dynamic-range extremes (RoadRunner and River Raid) still gives $r = 0.84$ ($n = 6$). *Decoder:* the predictor is about $L-F$ vs $T-F$, not L vs T , so the Seaquest decoder-fragility finding (§4.3) does not touch it—Seaquest stays in the upper-right helps-quadrant under both decoders ($T-F, L-F = (+22, +38)$ greedy, $(+80, +72)$ sampling), if anything reinforcing the relationship. What the decoder moves is the *secondary* latent-vs-text tally, not the task-level predictor.